

Surquaternions

Algebra, Infinitesimal Geometry,
and Support-Controlled Analysis

September 21, 2026

Abstract

A surquaternion is an element of $\mathbb{H}_{\mathbf{No}} = \mathbf{No} \oplus \mathbf{Noi} \oplus \mathbf{Noj} \oplus \mathbf{Nok}$ with Hamilton's multiplication. The definition is elementary; a usable theory must also say which parts of quaternionic geometry survive non-Archimedean scales, what an infinite sum denotes, and which notions of differentiation are compatible with the surreal field. The algebra is noncommutative, so factor order must be retained in every extension. Commutative results apply within explicitly identified central fields or complex slices; the quaternionic extensions require the additional arguments given here.

We develop the division algebra and its ordered-field-valued modulus, algebraically closed complex slices, conjugacy spheres, the algebraic rotation and spin description, one-sided polynomials with a constructive fundamental theorem and a complete zero-class algorithm, and finite-dimensional Hermitian spectral theory with a projector estimate at arbitrary surreal gap scale. We identify surquaternions with set-supported Hahn series having ordinary quaternion coefficients, prove the valuation, leading-coefficient and standard-part formulas with the *noncommutative* residue algebra $\mathbb{H}_{\mathbb{R}}$, and separate that summation theory from the exceptionally fine topology of the full surreal class. Positive support then supplies an exact infinitesimal exponential and logarithm, Baker–Campbell–Hausdorff at positive valuation, a filtered nonabelian rotation group with a graded bracket, a support-controlled multivariable implicit lifting theorem, and an exact square-root conditioning analysis. For analysis we keep slice regularity, Fueter regularity and field derivations apart, and prove coefficientwise representation, Cauchy and Fueter results on explicitly named coefficient spaces.

Two results are negative and sharp. The componentwise Berarducci–Mantova derivation admits no nonzero solution of $\partial_{\mathbb{H}}y = \mathbf{i}y$, so no global quaternionic exponential can satisfy the usual commuting chain rule for that derivation at $\omega\mathbf{i}$. And the global radial exponential built from the Ehrlich–Kaplan scalar normalization, whose complete logarithm fibres and four-coordinate differential we compute — finding rank-two critical spheres at radii in $\pi\mathbf{Oz}$, including the infinite radius ω — is not an additive-to-multiplicative homomorphism on noncommuting arguments. Three exponentials and four derivative-like operators are therefore fixed by name and symbol in section 2 and never identified.

Status. A self-contained mathematical development, with cited foundational inputs and proofs of the stated extensions. This is not a refereed paper, a priority claim, or a claim to solve a named published open problem. The accompanying symbolic checks verify finite identities, finite formal expansions and worked examples, not the general theorems. No Lean verification of this article is claimed.

Provenance. This is a merged report. It is the union of two independently written manuscripts, *noncommutative-hahn-obstructions* and *radial-exponential-support-lifting*, both dated September 21, 2026 and each delivered as a standalone archive with its own symbolic suite. Neither manuscript is reproduced here. Every result either manuscript reached is printed here; where the two proved the same statement by genuinely different routes, both routes are kept and marked; every honest limitation from either manuscript survives in section 17.

Repository baseline. The source repository is Vladimir Reshetnikov's `Surreal`, read at commit `39f2be6667ade51bca2b45daa47e289d69c09764`, accessed September 21, 2026. This article is a new standalone companion, not a modification of that repository.

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1 Scope, provenance, and the choice of analytic structure

1.1 What is being generalized

Hamilton’s quaternions do not require real completeness for their multiplication. They require an ordered field to make a sum of four squares positive, and a real closed field to obtain the square roots and complex factorizations used in much of their geometry. The surreal numbers supply both, while also supplying infinitely large and infinitesimal coefficients.

The basic object is therefore

$$\mathbb{H}_{\mathbf{No}} = \{a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k} : a, b, c, d \in \mathbf{No}\}, \quad \mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1. \quad (1.1)$$

We use *surquaternion* descriptively, without asserting that the word or the construction is new. A surquaternion is an ordered quadruple of surreals, not a new Conway cut with quaternionic left and right options. Throughout, the four coefficients are *central*: they commute with $\mathbf{i}, \mathbf{j}, \mathbf{k}$ and with every surquaternion. This is not an algebra of games with a new multiplication, nor an attempt to order the quaternion algebra. The order used to construct surreal numbers is an order on scalar components only; it does not become a compatible ordering of the division algebra (Proposition 3.3).

The one structural fact that governs the whole report is that $\mathbb{H}_{\mathbf{No}}$ is *noncommutative*. Conjugation, multiplicativity of the modulus, the meaning of “valuation”, leading coefficients, polynomial evaluation, matrix adjoints, exponential laws and every chain rule require separate treatment from the commutative surcomplex case. Where a companion report in the same repository proves a commutative statement, we identify its scalar or slice hypotheses before using it, and prove the additional noncommutative assertions over $\mathbb{H}$.

1.2 Three levels of scalars

Three levels organize the development, and a theorem at one level is never silently promoted to another.

Level	Available structure	What must not be inferred
$\mathbb{H}_F$, F real closed	Finite algebra, ordered-field-valued modulus, polynomial and matrix geometry	Real compactness, completeness, or unrestricted analytic transfer
D_Γ , a set-sized Hahn workspace	Well-ordered set-sized supports, valuation, exact Hahn evaluation	That every power series is evaluable everywhere, or that the workspace is closed under $\exp_{\mathbf{No}}$ or under a chosen derivation
$\mathbb{H}_{\mathbf{No}}$, the full class	All surreal scales and Conway normal forms	That ordinary convergent sequences model fine limits, or that class-indexed summation is available

Level 1 is finite algebra valid over any real closed field F ; this is the right level of generality, since the proofs then apply both to $\mathbf{No}$ and to suitable set-sized subfields. Level 2 adds well-ordered, set-sized supports and exact infinite summation. Level 3 consists of class statements. Analytic constructions additionally specify a domain, a support condition, and which differentiation is meant.

1.3 Relation to companion reports in the repository

Three committed reports in the same repository carry material that this one extends. We inherit and cite them rather than restating them, and we print only what noncommutativity changes.

Surcomplex Spectral Theory: Singular Values, Infinitesimal Rank, and Multiscale Stability, docs/surcomplex/spectral-theory/article.tex [15], already proves Hermitian and normal diagonalization, positive square roots, the singular value decomposition, polar decomposition and the variational principles with attained extrema over a real closed field and its complexification, on the stated ground that these need real closedness rather than Archimedean completeness; over a Hahn workspace it already has gap-sensitive spectral projectors and subspace perturbation, with an explicit warning about Gram squaring. Section 13 keeps the quaternionic case separate — the right-module structure, the adjoint $A^* = \bar{A}^T$ and the reality of eigenvalues of quaternionic Hermitian matrices are genuinely different — and cites that report for the complex case. One hard constraint is recorded at Warning 13.2: that report’s determinantal singular-scale theorem and its positive Cauchy–Binet argument use commutative minors and cannot be transferred unchanged to quaternionic matrices.

Trigonometry on the Surcomplex Plane: Canonical Phases, Arbitrary-Scale Geometry, and Infinitesimal Degeneration, docs/surcomplex/trigonometry/article.tex [14], establishes that every surcomplex direction has a *finite* surreal angle, and already contains the Ehrlich–Kaplan integer-part normalization with the omnific integers **Oz**, presenting it as one member of a character family and explicitly not claiming it as new. Sections 9 and 10 cite that report for the scalar normalization and its character-family caveat, and print only the conjugation-equivariant quaternionic radial extension, its full logarithm fibres, its four-coordinate differential and its rank-two critical points.

Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers, docs/surcomplex/differential-equations/article.tex [16], proves “the missing oscillator” three independent ways, with the consequence that $\partial^2 y + y = 0$ has only $y = 0$ and that $\sin \omega$ is not a Berarducci–Mantova differential element. Theorem 12.5 is the componentwise quaternionic analogue and is stated as an extension of that result, not as an independent discovery. That report also fixes a three-symbol phase convention precisely because three different objects were being called “phase”: the *finite angle* θ , the *infinitesimal phase residue* η , and the *accumulated phase* B . We reuse those three symbols in exactly those senses and introduce no fourth sense of the word; section 2 is written as an extension of that table.

A fourth companion, the physics report *Surreal Scalars and Spacetime* [17], asserts the group-homomorphism property of the *commutative* surcomplex Ehrlich–Kaplan exponential with kernel $2\pi\mathbf{iOz}$. Section 2.1 explains why that claim and the failure of the homomorphism law recorded in theorem 10.4 are both true, of different objects.

1.4 Foundational inputs and set-sized workspaces

We use the established facts that **No** is a real closed ordered class field, that every surreal has a unique Conway normal form with real coefficients and set-sized reverse-well-ordered exponents, and that Gonshor’s exponential is an ordered exponential-field operation [1, 2, 4, 6]. No proof here uses a sum indexed by a proper class.

Class language can be interpreted in a class theory such as NBG; in ZFC the same notation denotes definable class schemes. In a dependent type theory an ordinary small type cannot contain every ordinal and hence every surreal, so one works in a fixed universe or in a set-sized workspace and records the embeddings between workspaces. For the full surreal class, topological statements use a displayed ball basis, not a postulated set of all open subclasses. Algebraic group points are finite coordinate tuples. Statements about arbitrary class automorphisms or derivations concern individual class maps; we do not postulate a class whose elements are all class functions.

Write

$$t^\gamma := \omega^{-\gamma}, \quad (1.2)$$

the *Conway monomial map*. It must not be replaced, for arbitrary surreal γ , by a presumed identity with $\exp(-\gamma \log \omega)$ involving Gonshor's exponential. The two notational uses of powers are kept distinct throughout.

For a set-sized ordered subgroup $\Gamma \subset (\mathbf{No}, +)$ let

$$K_\Gamma = \mathbb{R}((t^\Gamma)), \quad D_\Gamma = \mathbb{H}_{K_\Gamma}.$$

Here K_Γ denotes the *real* Hahn field (the workspace denoted F_Γ in the repository notation guide), not its complexification. A Hahn series has a well-ordered support in Γ , with our increasing-exponent convention. If Γ is divisible, the standard real-closed Hahn-field theorem makes K_Γ real closed [3, 6]. That theorem, and the surreal normal-form theorem that embeds these series in $\mathbf{No}$, are foundational inputs, not consequences of quaternion multiplication.

Theorem 1.1 (Set localization). *Every set A of surquaternions is contained in some D_Γ with Γ a divisible, set-sized ordered subgroup of $(\mathbf{No}, +)$. Its elements are represented by their actual Conway normal forms, not merely by an abstract field embedding.*

Proof. Take the union of the supports of all four coordinates of all elements of A . This is a set of surreal exponents. Its rational linear span is a set-sized divisible subgroup Γ . All the coordinate series belong to K_Γ , so all the original quaternions belong to D_Γ . The normal-form addition and multiplication identify the operations there with those in $\mathbb{H}_{\mathbf{No}}$. $\square$

This localization does *not* assert that K_Γ is stable under the exponential, under a chosen surreal derivation, or under every analytic operation later considered; specified images can be placed in a larger workspace, and such closure is an additional hypothesis wherever those operations are used. Nor does localization identify the workspace's intrinsic topology with its subspace topology in the full surreal class: by theorem 7.3 these are dramatically different.

For fixed finite matrix sizes and polynomial degrees, algebraic statements about $\mathbb{H}_F$ can be translated into ordered-field statements about four coordinates per variable, so first-order real-closed-field transfer is available for such statements, subject to first-order expressibility [10]. It is not a license to transfer compactness, infinite summation, arbitrary function spaces, or the proper-class assertions about $\mathbf{No}$. Most proofs below avoid even this shortcut.

1.5 What is classical and what is developed here

Much of sections 3 to 5 is classical quaternion algebra over a real closed field [7], written out to make its hypotheses visible; its specialization to $\mathbf{No}$ does not make the division formula, the conjugacy classification or the spectral theorem new. The polynomial star product, spherical zero sets, the stem-function construction and the Fueter transformation belong to the established theory of quaternionic slice regularity [8]. The scalar global phase normalization of section 10.2 is due to Ehrlich and Kaplan [6] and is not introduced here.

The genuinely surreal content is the class/set boundary, exact support control, multiscale estimates, and the incompatibilities between otherwise natural analytic structures. Particularly useful combined results are set localization, the noncommutative residue calculus, the filtered rotation calculus, the complete zero-class algorithm, the spectral-projector estimate at arbitrary surreal gap

scale, support-certified implicit lifting, the square-root conditioning formula, the differential and logarithm fibres of the global radial exponential, and the no-oscillation theorem with its chain-rule corollary. These are presented as results proved in this article, not as exhaustively verified claims of first publication. The literature review is targeted, not a complete bibliographic classification.

2 The ledger: three exponentials, four derivatives, one name each

Read this section before any later one. Three exponentials and four derivative-like operators in this subject carry overlapping names in the literature and in the two source manuscripts, and each of them satisfies a different law on a different domain. Printing one “exponential” and one “derivative” would manufacture contradictions between statements that are all true. This section fixes one name and one symbol per object and records, for each law, exactly which object satisfies it.

Convention 2.1 (The three exponentials). (a) The *infinitesimal Hahn exponential* $\text{Exp}_\mathfrak{m}$, defined on the maximal ideal $\mathfrak{m}_H$ by the strong Hahn sum of $X^n/n!$, with inverse $\text{Log}_\mathfrak{m}$ on $1 + \mathfrak{m}_H$ (theorem 8.1). It is a bijection, it commutes with conjugation and with inner automorphisms, and it is additive *only on commuting pairs*; on noncommuting pairs the group law is BCH (Proposition 8.2). Source 10 writes $\text{Exp}_\mathfrak{m}$ and source 11 writes exp_H for this same object.

(b) The *finite-angle tube exponential* $\text{Exp}_\mathcal{T}$, defined on the tube $\mathcal{T} = \{a + v : |v| \text{ finite}\}$ as Gonshor’s real exponential times canonical finite-angle trigonometry (equation (10.1)). Its domain contains every finite quaternion and every central surreal but *excludes* $\omega\mathbf{i}$. It is *not* the strong Hahn sum of $q^n/n!$, for two distinct reasons (Warning 10.2), and it is not $\text{exp}(q_0) \text{Exp}_\mathfrak{m}(\varepsilon)$ in general. Source 10 writes $\text{Exp}_\mathfrak{p}$; source 11 does not isolate this intermediate object.

(c) The *global radial exponential* Exp_{rad} , defined on all of $\mathbb{H}_{\mathbf{No}}$ by equation (10.6) using the Ehrlich–Kaplan initial-integer-part normalization of the scalar phase, applied to the *actual modulus and direction* of the imaginary part. It is surjective onto $\mathbb{H}_{\mathbf{No}}^\times$, conjugation-equivariant, and obeys the addition law on commuting pairs; it is *not* an everywhere Hahn-summed power series and *not* an additive-to-multiplicative homomorphism on the whole noncommutative algebra, so its fibre over 1 must not be called a kernel (theorems 10.4 and 10.5). Source 10 writes Exp_{EK} and source 11 writes Exp_{can} for this same object.

The scalar Gonshor exponential on $\mathbf{No}$ is written $\text{exp}_{\mathbf{No}}$, with inverse $\log_{\mathbf{No}}$; it does obey the unrestricted additive-to-multiplicative law, on the commutative field $\mathbf{No}$.

Convention 2.2 (The four derivative-like operators). (a) $\partial_\mathbb{H}$, the *componentwise Berarducci–Mantova derivation*: the scalar derivation ∂ on $\mathbf{No}$, normalized by $\partial\omega = 1$ and extended coordinatewise with $\partial_\mathbb{H}\mathbf{i} = \partial_\mathbb{H}\mathbf{j} = \partial_\mathbb{H}\mathbf{k} = 0$, so that it differentiates the surreal coefficients themselves. Its constant algebra is exactly $\mathbb{H}_\mathbb{R}$ and it is surjective (Corollary 12.2). Source 10 writes D and source 11 writes ∂_H . More generally $\delta_\mathbb{H}$ denotes the componentwise extension of an *arbitrary* scalar field derivation δ (theorem 12.1), and ad_w the inner part of a general derivation.

(b) The *coefficientwise slice derivative* ∂_{sl} , acting on a right-coefficient series by $\partial_{\text{sl}}(\sum_n q^n a_n) = \sum_{n \geq 1} n q^{n-1} a_n$ (Propositions 14.1 and 14.4). This is a formal or slice derivative in the *variable*; it is not the full algebraic differential, and it is not $\partial_\mathbb{H}$.

(c) The *fine differential* df_q , the four-coordinate $\mathbf{No}$ -linear Fréchet derivative of Definition 11.1, used to compute $d\text{Exp}_{\text{rad}}$ in theorem 11.2. Even the polynomial q^2 has fine differential

$h \mapsto qh + hq$ rather than $h \mapsto 2hq$ (Proposition 15.2). A nonsingular fine differential is never used here as an inverse function theorem.

- (d) The *left Fueter operator* $\mathcal{D} = \partial_a + \mathbf{i}\partial_{x_1} + \mathbf{j}\partial_{x_2} + \mathbf{k}\partial_{x_3}$, with the four-coordinate Laplacian Δ_4 (section 14.5). Here $\partial_a, \partial_{x_m}, \partial_r, \partial_b$ are *variable* derivatives of ordinary coefficient functions and their Taylor extensions; they are never $\partial_{\mathbb{H}}$. The identity map is slice regular but has $\mathcal{D}q = -2$, so slice regularity does not imply Fueter regularity.

Reusing the three phase symbols of [16]: θ always denotes a *finite* angle (Proposition 9.1); η always denotes an infinitesimal phase residue, so that $\theta = \text{st}(\theta) + \eta$; and B denotes an accumulated phase, an arbitrary surreal with $\partial B = b$, which need not be finite and is not the angle of anything when it is infinite. No fourth sense of “phase” is introduced, and Sin, Cos of equation (10.4) are the global integer-part-normalized functions, distinct from the finite-angle $\sin, \cos$ of equation (9.1).

Object	Domain	Laws it satisfies	Laws it does <i>not</i> satisfy
$\text{Exp}_{\mathfrak{m}}, \text{Log}_{\mathfrak{m}}$	$\mathfrak{m}_H \leftrightarrow 1 + \mathfrak{m}_H$, by strong Hahn summation	Bijection; conjugation- and inner-equivariant; $w(\text{Exp}_{\mathfrak{m}}(x) - 1) = w(x)$; additive on commuting pairs; BCH otherwise	Not additive on noncommuting pairs; not an additive-group isomorphism onto $1 + \mathfrak{m}_H$, the transported law being noncommutative
$\text{Exp}_{\mathcal{T}}$	the tube $\mathcal{T}$, excluding $\omega \mathbf{i}$	Conjugation-equivariant; $ \text{Exp}_{\mathcal{T}}(a + v) = e^a$; addition law on commuting pairs; surjective onto $\mathbb{H}_{\mathbf{No}}^{\times}$; polar logarithms and n th roots	Not a strong sum of $q^n/n!$; not $\exp(q_0) \text{Exp}_{\mathfrak{m}}(\varepsilon)$; no preferred logarithm on the negative scalar axis
Exp_{rad}	all of $\mathbb{H}_{\mathbf{No}}$, onto $\mathbb{H}_{\mathbf{No}}^{\times}$	Conjugation-equivariant and inverse-compatible, with $\ \text{Exp}_{\text{rad}} q\ = \exp_{\mathbf{No}}(\text{Re } q)$; agrees with $\text{Exp}_{\mathfrak{m}}$ on $\mathfrak{m}_H$; on each slice a surjective homomorphism with kernel $2\pi \mathbf{Oz}I$; multiplicative on commuting pairs	Not an everywhere Hahn-summed power series; <i>not</i> an additive-to-multiplicative homomorphism on noncommuting arguments; fibre over 1 is not a kernel; no commuting chain rule for $\partial_{\mathbb{H}}$
$\partial_{\mathbb{H}}$	all of $\mathbb{H}_{\mathbf{No}}$, coefficient-wise	Leibniz; strongly additive; constant algebra $\mathbb{H}_{\mathbb{R}}$; surjective; $\partial_{\mathbb{H}}(\mathcal{O}_H) \subseteq \mathfrak{m}_H$	No solution of $\partial_{\mathbb{H}}y = cIy$ or ycI ; hence no global commuting exponential chain rule (Corollary 12.6)
$\delta_{\mathbb{H}}, \text{ad}_w$	$\mathbb{H}_F$, any scalar derivation δ	$D = \delta_{\mathbb{H}} + \text{ad}_w$ uniquely; $\delta N(q) = 2 \text{Re}(\delta_{\mathbb{H}}(q)\bar{q})$	$\delta_{\mathbb{H}}(q^n) = nq^{n-1}\delta_{\mathbb{H}}q$ only under commutation
∂_{sl}	admissible right-coefficient series	Preserves the common-support class; coefficient-wise Cauchy–Riemann identities	Not the full differential: $d(q^2)_q \neq h \mapsto 2hq$
df_q	class scheme on $\mathbb{H}_{\mathbf{No}}$	$\mathbf{No}$ -linear; computes $d \text{Exp}_{\text{rad}}$ and $\det d \text{Exp}_{\text{rad}}$	No Banach structure; no inverse function theorem; zero differential does not imply constancy (Example 7.5)
$\mathcal{D}, \Delta_4$	coefficient functions in four variables	$\mathcal{D}\Delta_4 f = 0$ for coherent slice-regular f and for slice polynomials	$\mathcal{D}f = 0$ does not follow from slice regularity: $\mathcal{D}q = -2$

2.1 Why the homomorphism claim and its failure are both true

The physics companion [17] derives, for the *commutative* surcomplex field $\mathbf{No}[\mathbf{i}]$, the identity $\text{Exp}_{\text{EK}}(a + \mathbf{i}b) = \exp_{\mathbf{No}}(a)(\text{Cos } b + \mathbf{i} \text{Sin } b)$ and states that it extends the ordinary complex exponential and *has* the group-homomorphism property, with kernel $2\pi \mathbf{iOz}$. Theorem 10.4 states that Exp_{rad} is *not* an additive-to-multiplicative homomorphism on noncommuting arguments. These two statements are consistent, and the reason is structural rather than a matter of emphasis.

Restrict Exp_{rad} to one complex slice $\mathbf{No} + \mathbf{No}I$. That slice is a commutative field isomorphic to $\mathbf{No}[\mathbf{i}]$ (theorem 4.1), the restriction *is* the surcomplex Exp_{EK} by theorem 10.4, and on it the homomorphism law holds with kernel exactly $2\pi \mathbf{Oz}I$. The physics report’s claim is a claim about

that commutative object, and it is correct. The global failure is a failure at *noncommuting* arguments p, q : there is then no slice containing both, so $\text{Exp}_{\text{rad}}(p + q)$ is computed from the modulus and direction of $\text{Im}(p + q)$, which bear no multiplicative relation to those of $\text{Im } p$ and $\text{Im } q$. Example 10.6 is the exact witness: $\text{Exp}_{\text{rad}}(\omega \mathbf{i} + \pi \mathbf{j}) = 1 + \frac{\pi^2}{2\omega} \mathbf{i} + \text{O}(\omega^{-2})$ while $\text{Exp}_{\text{rad}}(\omega \mathbf{i}) \text{Exp}_{\text{rad}}(\pi \mathbf{j}) = -1$. Both computations are exact, and the two arguments $\omega \mathbf{i}$ and $\pi \mathbf{j}$ do not commute. So the homomorphism law is a theorem on each slice and false across slices, and the additive group being abelian while the multiplicative group is not already shows that a global homomorphism is not available.

Commutation is a sufficient condition for this law, not a necessary condition for an individual pair: $p = 6\pi \mathbf{i}$, $q = 8\pi \mathbf{j}$ do not commute, but $\text{Exp}_{\text{rad}}(p) = \text{Exp}_{\text{rad}}(q) = \text{Exp}_{\text{rad}}(p + q) = 1$, since $|p + q| = 10\pi$.

A third statement must be kept out of this comparison. Corollary 12.6 says that no map $E : \mathbb{H}_{\mathbf{No}} \rightarrow \mathbb{H}_{\mathbf{No}}^{\times}$ satisfies $\partial_{\mathbb{H}}(E(q)) = E(q)\partial_{\mathbb{H}}q$ on commuting pairs $q, \partial_{\mathbb{H}}q$. That is a differential-algebraic statement about the specific derivation $\partial_{\mathbb{H}}$, not a statement about the multiplicative behaviour of any exponential. It is compatible with Exp_{rad} being a slicewise homomorphism, and it is compatible with theorem 11.2, which computes the *fine* differential $d\text{Exp}_{\text{rad}}$ — a different operator again, in the sense of Convention 2.2(c). All three statements hold, of three different objects, and flattening any two of them into one would manufacture a false contradiction across this collection.

3 The division algebra and its norm

Let F initially be any ordered field. Write $q = (a, u) = a + u$ with $a \in F$ and $u \in F^3$ the pure imaginary part, identified with the span of $\mathbf{i}, \mathbf{j}, \mathbf{k}$. The multiplication law is

$$(a + u)(b + v) = ab - u \cdot v + av + bu + u \times v, \quad (3.1)$$

where the dot and cross products are the polynomial formulas with the usual orientation $(\mathbf{i}, \mathbf{j}, \mathbf{k})$. No metric completeness is hidden in this notation. Define

$$\bar{q} = a - u, \quad T(q) = q + \bar{q} = 2a, \quad \text{Re } q = a, \quad \text{Im } q = u, \quad N(q) = q\bar{q} = a^2 + u \cdot u.$$

The notation $\text{Re } q$ means the *central scalar part*; it need not be an ordinary real number. Quaternion conjugation is distinct from a derivation, from the standard part map of theorem 6.3, and from conjugation inside a chosen complex slice.

Proposition 3.1 (Associativity and change of scalars). *$\mathbb{H}_F$ is an associative unital F -algebra of dimension four. Every ordered field embedding $F \hookrightarrow E$ induces an injective algebra homomorphism $\mathbb{H}_F \hookrightarrow \mathbb{H}_E$ preserving conjugation and norm square.*

Proof. Impose Hamilton's relations in the free associative algebra and reduce every word to the span of $1, \mathbf{i}, \mathbf{j}, \mathbf{k}$; independence follows from the faithful matrix representation Proposition 4.3. Alternatively equation (3.1) verifies associativity directly as a polynomial identity in twelve scalar variables, which is one of the recorded symbolic checks. Applying the scalar embedding to each of four coordinates proves the last assertion. $\square$

Theorem 3.2 (Hamilton algebra over an ordered field). *The algebra $\mathbb{H}_F$ is an associative division algebra with center exactly F . Conjugation is an anti-involution, $\overline{pq} = \bar{q}\bar{p}$, and*

$$q\bar{q} = \bar{q}q = N(q), \quad N(pq) = N(p)N(q), \quad q^{-1} = \frac{\bar{q}}{N(q)} \quad (q \neq 0). \quad (3.2)$$

Every element satisfies

$$q^2 - T(q)q + N(q) = 0. \quad (3.3)$$

Proof. The conjugation identity follows from the relations on the basis, or from equation (3.1), and gives $q\bar{q} = \bar{q}q = N(q)$. If $q \neq 0$ then at least one coordinate square is positive, and a sum of squares in an ordered field vanishes only when every summand does, so $N(q) > 0$; the inverse formula follows. Consequently $N(pq) = pq\bar{q}\bar{p} = N(q)p\bar{p} = N(p)N(q)$. An element commuting with both $\mathbf{i}$ and $\mathbf{j}$ has zero imaginary coordinates, so the center is exactly F . Expanding $q(q + \bar{q})$ proves equation (3.3). $\square$

In particular $\mathbb{H}_{\mathbf{No}}$ has no zero divisors. Its ordered geometry comes from F , not from an order on $\mathbb{H}_F$ itself.

Proposition 3.3 (Center, commutators, and the absence of a ring order). *The center of $\mathbb{H}_F$ is F , and*

$$[a + u, b + v] = 2u \times v, \quad N([p, q]) \leq 4N(\text{Im } p)N(\text{Im } q).$$

There is no ordered-division-ring ordering on $\mathbb{H}_F$ extending the order of F .

Proof. Equation (3.1) gives the commutator, and a vector whose cross product with each coordinate vector vanishes is zero, which is the assertion about the center. The identity $N(u \times v) = N(u)N(v) - (u \cdot v)^2$ proves the squared-norm inequality, which requires no square roots in F . In an ordered division ring every nonzero square is positive, contradicting $\mathbf{i}^2 = -1$. $\square$

Suppose from now on that F is real closed, unless a weaker hypothesis is stated. Define the F -valued modulus and the scalar inner product by

$$|q| = \sqrt{N(q)}, \quad (p, q) = \text{Re}(p\bar{q}).$$

Taking positive square roots in Proposition 3.3 now gives $|[p, q]| \leq 2|\text{Im } p||\text{Im } q|$. The distinction matters already over $F = \mathbb{Q}$: $N(1 + \mathbf{i}) = 2$ belongs to F , whereas its positive square root does not. This is not an ordinary real norm when F is non-Archimedean, and an implementation must not replace it by a machine floating-point norm.

Proposition 3.4 (Euclidean identities without completeness). *For $p, q \in \mathbb{H}_F$,*

$$|pq| = |p||q|, \quad |p + q| \leq |p| + |q|, \quad |(p, q)| \leq |p||q|, \quad \max_m |q_m| \leq |q| \leq 2 \max_m |q_m|.$$

For nonzero p, q , equality in the triangle inequality holds exactly when $q = \lambda p$ for some $\lambda \in F_{>0}$. Moreover, for invertible p, q ,

$$p^{-1} - q^{-1} = p^{-1}(q - p)q^{-1}, \quad |p^{-1} - q^{-1}| = \frac{|p - q|}{|p||q|}. \quad (3.4)$$

Finally $\mathbb{H}_F \otimes_F F[\mathbf{i}] \simeq M_2(F[\mathbf{i}])$, a splitting that introduces zero divisors only after extension.

Proof. Multiplicativity and the coordinate bounds follow from multiplicativity of N and the four coordinate squares. For Cauchy–Schwarz, the polynomial $N(p - sq)$ in the central variable s is nonnegative for every $s \in F$. The case $q = 0$ is immediate; otherwise substituting $s = (p, q)/N(q)$ gives $(p, q)^2 \leq N(p)N(q)$. Therefore $N(p + q) \leq (|p| + |q|)^2$, proving the triangle inequality; equality forces $(p, q) = |p||q|$ and equality in Cauchy–Schwarz, and the same substitution gives

$p = ((p, q)/N(q))q$ with positive scalar coefficient. The inverse identity is verified by multiplication, and its modulus is exactly multiplicative by the first assertion. No supremum, compactness or completeness is used. The splitting is Proposition 4.3. $\square$

Example 3.5 (Several scales in one number). For $q = \omega + \omega^{-1}\mathbf{i} + \mathbf{j} + \omega^{-\omega}\mathbf{k}$,

$$N(q) = \omega^2 + 1 + \omega^{-2} + \omega^{-2\omega}, \quad q^{-1} = \frac{\omega - \omega^{-1}\mathbf{i} - \mathbf{j} - \omega^{-\omega}\mathbf{k}}{\omega^2 + 1 + \omega^{-2} + \omega^{-2\omega}}.$$

The coordinates may have arbitrarily different orders of magnitude without compromising associativity, division, or the norm identities.

Remark 3.6 (Which quaternion symbol algebra this is). A quaternion symbol algebra $(a, b)_F$ over a real closed field is split unless both a and b are negative: a positive parameter is a square and produces a nontrivial idempotent, whereas two negative parameters can be rescaled to $(-1, -1)_F$. Thus Hamilton's algebra is the quaternion division-algebra case. This modest classification suffices here; no classification of arbitrary class-sized division algebras is invoked.

4 Complex slices, conjugacy, rotations, and rational charts

Define the sphere of imaginary units

$$\mathbb{S}(F) = \{I \in \mathbb{H}_F : \operatorname{Re} I = 0, N(I) = 1\},$$

which by equation (3.3) is exactly the solution set of $I^2 = -1$. For $I \in \mathbb{S}(F)$ the *slice* $F_I = F + FI$ is isomorphic to $F[\mathbf{i}]$ and is therefore algebraically closed when F is real closed. Slices intersect in F unless their units differ by sign; their union is the whole algebra, but not a disjoint union, and a product of elements of two different slices need not lie in either.

Theorem 4.1 (Slice decomposition, centralizers, conjugacy). *Let F be real closed. Every noncentral q has a unique expression*

$$q = a + rI, \quad a = \operatorname{Re} q, \quad r = |\operatorname{Im} q| > 0, \quad I = \frac{\operatorname{Im} q}{|\operatorname{Im} q|} \in \mathbb{S}(F), \quad (4.1)$$

its centralizer is exactly F_I , and its minimal polynomial over F is $X^2 - 2aX + (a^2 + r^2)$. Two quaternions are conjugate by an invertible quaternion if and only if they have the same trace and norm; the conjugacy class of $a + rI$ is

$$[a, r] = a + r\mathbb{S}(F),$$

and the conjugator may be taken of modulus one. Every commuting pair lies in a common slice.

Proof. The commutator of (a, u) and (b, v) is $(0, 2u \times v)$; for $u \neq 0$ its vanishing is equivalent to v being an F -multiple of u , which proves the centralizer assertion and the last sentence, any slice being available for two central elements. The quadratic follows from $I^2 = -1$, and no degree-one central polynomial annihilates a noncentral element. Conjugation preserves scalar part and norm. Conversely let $I, J \in \mathbb{S}(F)$. If $J \neq -I$ then $h = 1 - JI \neq 0$ satisfies $hI = I + J = Jh$. If $J = -I$, choose a unit imaginary K perpendicular to I ; then $KI = -IK = JK$. Normalize h by its positive modulus. $\square$

A nonreal surquaternion thus has a proper-class sphere of conjugates even though its minimal polynomial over the center has degree two. Polynomial degree bounds must count conjugacy classes or multiplicities, never individual points on such a sphere.

Proposition 4.2 (Explicit square roots). *For $q = a + v \neq 0$ put $R = |q|$. If q is not a nonpositive central scalar, let $s_0 = \sqrt{(R + a)/2} > 0$; then the two square roots of q are*

$$\pm \left(s_0 + \frac{v}{2s_0} \right). \quad (4.2)$$

A negative central scalar $-\rho$, $\rho > 0$, has the roots $\sqrt{\rho} I$ for every $I \in \mathbb{S}(F)$. A positive central scalar has exactly its two central square roots, and zero has only zero. No “infinite angle” is required to solve these equations.

Proof. If $v \neq 0$ then $R > |a|$. The square of equation (4.2) has imaginary part v and scalar part $s_0^2 - |v|^2/(4s_0^2) = a$, using $|v|^2 = R^2 - a^2$. Any square root commutes with its square, so for noncentral q it lies in F_I , where a quadratic has only two roots. For central q , writing a prospective root $b + u$ gives imaginary part $2bu$: if $u = 0$ one obtains the central roots, and if $b = 0$ its square is $-|u|^2$, giving exactly the negative-scalar sphere. Positivity treats the remaining cases. $\square$

For example $2 + \mathbf{i}$ squares to $3 + 4\mathbf{i}$, while $\omega^{-1}\mathbf{i} + \omega^{-2}\mathbf{j}$ has a square-root scale determined by the positive square root of its norm.

4.1 Matrix realization

Put $C_F = F[\mathbf{i}]$ and write $q = z + w\mathbf{j}$ with $z, w \in C_F$. Since $\mathbf{j}z = \bar{z}\mathbf{j}$,

$$(z + w\mathbf{j})(z' + w'\mathbf{j}) = (zz' - w\bar{w}') + (zw' + w\bar{z}')\mathbf{j}.$$

Proposition 4.3 (Matrix realization). *The map*

$$\rho(z + w\mathbf{j}) = \begin{pmatrix} z & w \\ -\bar{w} & \bar{z} \end{pmatrix} \quad (4.3)$$

is a faithful algebra homomorphism $\mathbb{H}_F \rightarrow M_2(C_F)$ with $\rho(\bar{q}) = \rho(q)^$, $\det \rho(q) = N(q)$ and $\text{tr } \rho(q) = T(q)$. The modulus-one group is identified with $\text{SU}(2, C_F)$, and scalar extension splits the algebra, $\mathbb{H}_F \otimes_F C_F \simeq M_2(C_F)$.*

Proof. The displayed product law is precisely matrix multiplication in equation (4.3); the determinant, trace and star formulas follow directly. A unitary 2×2 matrix of determinant one has the displayed form, by comparing its adjugate with its conjugate transpose. The four images of the basis elements are linearly independent over C_F and span the four-dimensional matrix algebra. $\square$

The image of $\mathbb{H}_F$ is a special four-dimensional real form of the matrix algebra, not the entire matrix algebra. Complex scalar extension splits the algebra and introduces zero divisors; it does not exhibit zero divisors in $\mathbb{H}_F$.

4.2 The rotation double cover as an algebraic statement

Let $V = \text{Im } \mathbb{H}_F \simeq F^3$ and $S^3(F) = \{u \in \mathbb{H}_F : N(u) = 1\} = \text{Sp}(1, F)$. Every $q \neq 0$ has the unique polar factorization $q = |q|u$ with $u \in S^3(F)$, and because the positive scalar factor is central,

$$\mathbb{H}_F^\times \simeq F_{>0} \times S^3(F) \quad (4.4)$$

as groups. This is algebraic polar decomposition; it does not assert topological compactness of $S^3(\mathbf{No})$. For a quaternion $u = a + v$,

$$u x \bar{u} = (a^2 - |v|^2)x + 2(v \cdot x)v + 2a(v \times x) \quad (x \in V), \quad (4.5)$$

an exact polynomial identity at arbitrary surreal scales, requiring no unit hypothesis; for $N(u) = 1$ it equals uxu^{-1} . Writing $u = c + sI$ with $c^2 + s^2 = 1$ turns equation (4.5) into the unnormalized Rodrigues form $(c^2 - s^2)x + 2s^2(I \cdot x)I + 2cs(I \times x)$.

Theorem 4.4 (Rotations, spin, and inner automorphisms). *Over a real closed field F , the map $u \mapsto R_u = (x \mapsto uxu^{-1})$ gives an exact sequence of groups*

$$1 \longrightarrow \{1, -1\} \longrightarrow S^3(F) \longrightarrow \text{SO}(3, F) \longrightarrow 1, \quad (4.6)$$

and

$$\text{Aut}_F(\mathbb{H}_F) \simeq \text{SO}(3, F) \simeq \mathbb{H}_F^\times / F^\times.$$

In particular every center-fixing automorphism of the surquaternions is inner. Moreover

$$S^3(F) \times S^3(F) \longrightarrow \text{SO}(4, F), \quad (u, v) : q \longmapsto uqv^{-1}, \quad (4.7)$$

is surjective with kernel $\{(1, 1), (-1, -1)\}$.

Proof. Conjugation preserves V , its norm and its cross product, so its matrix lies in $\text{SO}(3, F)$; its kernel commutes with every imaginary quaternion, hence is central, and central elements of modulus one are ± 1 .

Surjectivity, first route. Let $A \in \text{SO}(3, F)$. Orthogonality and determinant one give $\det(A - I) = -\det(A + I)$, so A has a nonzero fixed vector; normalize it to $I \in V$. On $I^\perp$ the transformation is a two-dimensional rotation with entries $c, s \in F$, $c^2 + s^2 = 1$, in an appropriately oriented orthonormal basis. If $c \neq -1$ take $a = \sqrt{(1+c)/2}$ and $b = s/(2a)$, so $a^2 + b^2 = 1$ and equation (4.5) shows that $a + bI$ induces A ; if $c = -1$, use I itself.

Surjectivity, second route. The cubic characteristic polynomial of A has a root in F , and every orthogonal eigenvalue in F is 1 or -1 . If only a -1 root is found, its perpendicular plane has determinant -1 and hence eigenvalues $1, -1$, since its characteristic polynomial has discriminant $\text{tr}^2 + 4 > 0$. Either way a fixed vector exists; finish as before. The two arguments are independent; the first is the shorter, the second avoids the determinant identity and only uses that an odd-degree polynomial over a real closed field has a root.

For the automorphism statement, an F -algebra automorphism preserves the coefficients of the quadratic minimal polynomial of every noncentral element, hence scalar part and norm, hence the imaginary dot and cross products by equation (3.1); its action on V lies in $\text{SO}(3, F)$. Conversely each such orthogonal transformation preserves equation (3.1) and extends by fixing F ; surjectivity makes it inner. Equivalently, an automorphism sends $\mathbf{i}, \mathbf{j}, \mathbf{k}$ to an oriented orthonormal imaginary triple, and every such triple preserves Hamilton multiplication.

For equation (4.7), take an orientation-preserving isometry A of $\mathbb{H}_F$ and put $z = A(1)$. Left multiplication by z^{-1} has determinant 1 and makes the composite fix 1; its restriction to V lies in $\mathrm{SO}(3, F)$, so equals conjugation by a unit u , and $A(q) = zuqu^{-1}$. Here the determinant of left multiplication by q is $N(q)^2$, obtainable from its four-coordinate matrix. For the kernel, $uqv^{-1} = q$ for every q first gives $u = v$ at $q = 1$, then forces u to be central. A central modulus-one scalar is 1 or -1 , so the kernel is exactly $\{(1, 1), (-1, -1)\}$. $\square$

Equation (4.6) is a two-to-one map on algebraic group points. It is *not*, without a separate topology and definition, a covering-space theorem for the fine surreal topology of section 7. An arbitrary ring automorphism may also act on the center: restriction to F and the coefficientwise extension of a field automorphism show that for set-sized F the full automorphism group is a semidirect product of the inner automorphisms with $\mathrm{Aut}(F)$. For **No** this is read for each individual class automorphism, without forming a collection of all class functions.

Consequently arbitrary-scale spatial rotations need no trigonometric value at an infinite argument: they are constructed by rational operations and positive square roots alone. The source manuscripts give Cayley and stereographic presentations of the same rational chart on the unit group. Both formulas are exact at every scale, including infinitesimal ones.

Proposition 4.5 (Cayley and stereographic presentations of one chart). *For $x \in V$ the Cayley chart*

$$\mathrm{Cay}(x) = (1 + x)(1 - x)^{-1} \quad (4.8)$$

is defined because $N(1 - x) = 1 + N(x) > 0$, has modulus one, never equals -1 , and is inverted on $S^3(F) \setminus \{-1\}$ by $x = (u - 1)(u + 1)^{-1}$, which is imaginary. Equivalently, its stereographic presentation

$$u = \frac{1 - |x|^2 + 2x}{1 + |x|^2}, \quad x = \frac{\mathrm{Im} u}{1 + \mathrm{Re} u} \quad (u \neq -1), \quad (4.9)$$

has positive denominators and exact modulus one.

Proof. Since $\bar{x} = -x$, conjugation reverses the two factors and gives

$$\overline{\mathrm{Cay}(x)} = (1 + x)^{-1}(1 - x) = \mathrm{Cay}(x)^{-1}.$$

All displayed rational functions of the single element x commute, and $N(1 + x) = N(1 - x) = 1 + N(x)$, proving modulus one. The equation $\mathrm{Cay}(x) = -1$ would give $1 + x = -1 + x$, which is impossible. Conversely, for $N(u) = 1$ and $u \neq -1$, let $x = (u - 1)(u + 1)^{-1}$. Using $\bar{u} = u^{-1}$ gives $\bar{x} = (u^{-1} + 1)^{-1}(u^{-1} - 1) = -x$; substitution recovers u . Thus this inverse is imaginary and the chart is a bijection onto the indicated domain.

Finally $(1 - x)^{-1} = (1 + x)/(1 + N(x))$ and $x^2 = -N(x)$ give

$$\mathrm{Cay}(x) = \frac{(1 + x)^2}{1 + N(x)} = \frac{1 - N(x) + 2x}{1 + N(x)},$$

which identifies the two presentations. Independently checking the stereographic norm amounts to $(1 - N(x))^2 + 4N(x) = (1 + N(x))^2$. Its inverse follows from $\mathrm{Im} u = 2x/(1 + N(x))$ and $1 + \mathrm{Re} u = 2/(1 + N(x)) > 0$. Neither presentation asserts that $\mathrm{Sp}(1, F)$ is topologically a real three-sphere. $\square$

4.3 Standard part of the unit group

Every coordinate of a unit surquaternion has absolute value at most 1, so every unit quaternion is finite in the sense of theorem 6.3. Standard part therefore gives a split exact sequence

$$1 \longrightarrow U_1 \longrightarrow S^3(\mathbf{No}) \xrightarrow{\text{st}} S^3(\mathbb{R}) \longrightarrow 1, \quad U_1 = S^3(\mathbf{No}) \cap (1 + \mathfrak{m}_H), \quad (4.10)$$

whose section is the inclusion of constant quaternions. The kernel U_1 describes infinitesimal rotations and is nonabelian; theorem 8.1 and theorem 8.5 identify it, as a set with an explicitly transported group law, with the infinitesimal imaginary quaternions. The corresponding statement over a workspace D_Γ reads $1 \rightarrow U_1 \rightarrow U \rightarrow \text{Sp}(1, \mathbb{R}) \rightarrow 1$ with $U = \{u \in \mathcal{O}_D : N(u) = 1\}$.

4.4 Quaternionic fractional linear transformations

A further exact geometry is the right projective line $\mathbb{P}^1(\mathbb{H}_F)$. A matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2(\mathbb{H}_F)$ acts by

$$f(q) = (aq + b)(cq + d)^{-1} \quad (4.11)$$

where the denominator is nonzero, with the usual projective interpretation at its pole and at infinity. Right lines are essential: the column $(q, 1)^T$ is multiplied on the left by M , then normalized on the right by $(cq + d)^{-1}$. For finite affine arguments in its domain, direct subtraction gives

$$(f(q) - f(p))(cp + d) = (a - f(q)c)(q - p), \quad (4.12)$$

so the algebraic differential is $df_q(h) = (a - f(q)c)h(cq + d)^{-1}$. Its first factor is nonzero, since otherwise the row (a, b) would equal $f(q)(c, d)$, contradicting invertibility of M . Consequently

$$|df_q(h)| = \frac{|a - f(q)c|}{|cq + d|} |h|,$$

so these transformations are conformal in this exact algebraic differential sense over F . No real compactification, manifold completeness or contour integration is needed. Formula equation (4.12) also shows why moving a denominator to the wrong side changes the transformation.

5 One-sided polynomials: roots, spheres, and multiplicity

A polynomial with quaternion coefficients needs an evaluation convention. We use a *central* indeterminate X with coefficients on the right:

$$P(X) = \sum_{n=0}^d X^n a_n \in \mathbb{H}_F[X], \quad P(q) = \sum_{n=0}^d q^n a_n.$$

The product in this ring is written $\star$ when it must be distinguished from pointwise multiplication of the evaluated functions, which it is not. Centrality of X does not make evaluation multiplicative. These conventions agree with the polynomial part of slice regularity [8].

Lemma 5.1 (Twisted product and left division). *If $P(q) \neq 0$ then*

$$(P \star Q)(q) = P(q) Q(P(q)^{-1} q P(q)), \quad (5.1)$$

while if $P(q) = 0$ then $(P \star Q)(q) = 0$. For any q there is a unique polynomial Q with

$$P = (X - q) \star Q + P(q), \quad (5.2)$$

so $P(q) = 0$ if and only if $P = (X - q) \star R$ for some R : a zero gives a left linear factor when coefficients are evaluated on the right.

Proof. Writing $Q = \sum_n X^n b_n$ and regrouping finite sums, the product evaluates to $\sum_n q^n P(q) b_n$. If $P(q) \neq 0$, use $q^n P(q) = P(q)(P(q)^{-1} q P(q))^n$; if $P(q) = 0$ every term vanishes. For division, solve coefficients downward: $b_{d-1} = a_d$ and $b_{n-1} = a_n + q b_n$, with final remainder $a_0 + q b_0 = \sum_n q^n a_n$. This also proves uniqueness. $\square$

Let $P^c = \sum X^n \bar{a}_n$ and define the *normal polynomial*, equivalently the central symmetrization,

$$N_P = P \star P^c = P^c \star P \in F[X]. \quad (5.3)$$

Writing $P = P_0 + P_1 \mathbf{i} + P_2 \mathbf{j} + P_3 \mathbf{k}$ with $P_m \in F[X]$, whose components commute, gives the explicit central form $N_P = P_0^2 + P_1^2 + P_2^2 + P_3^2$. For degree d its degree is $2d$ with leading coefficient $N(a_d) > 0$, and $N_{P \star Q} = N_P N_Q$. At a central x , $N_P(x) = N(P(x))$; at a noncentral quaternion one must not replace the polynomial normal by that pointwise formula.

For a noncentral class $[a, r]$, $r > 0$, set $\Delta_{a,r}(X) = X^2 - 2aX + a^2 + r^2$, which is irreducible over F and vanishes at every point of that class. Divide centrally:

$$P = \Delta_{a,r} Q + XA + B, \quad A, B \in \mathbb{H}_F. \quad (5.4)$$

Lemma 5.2 (Remainder on a sphere). *On the sphere $[a, r]$ evaluation is $P(q) = qA + B$. Reducing N_P modulo $\Delta_{a,r}$ gives the two central coefficients*

$$(2aN(A) + 2\operatorname{Re}(A\bar{B}))X + N(B) - (a^2 + r^2)N(A). \quad (5.5)$$

Hence P has a zero on $[a, r]$ if and only if $\Delta_{a,r} \mid N_P$, and then exactly one of the following holds: $A = B = 0$ and the entire sphere is a zero set; or $A \neq 0$ and the unique zero on that sphere is

$$q = -BA^{-1} = -B\bar{A}/N(A), \quad \operatorname{Re} q = a, \quad N(q) = a^2 + r^2. \quad (5.6)$$

Proof. The characteristic quadratic vanishes on the sphere, which gives the evaluation. Two independent routes complete the proof, and both were carried out in the source manuscripts.

Route one, via the norm of the remainder. Modulo $\Delta_{a,r}$ the normal polynomial is congruent to $N(XA + B) = N(A)X^2 + 2\operatorname{Re}(A\bar{B})X + N(B)$. If $\Delta_{a,r} \mid N_P$ and $A = 0$, this constant remainder forces $N(B) = 0$, hence $B = 0$. If $A \neq 0$, the degree-two polynomial must equal $N(A)\Delta_{a,r}$, so $\operatorname{Re}(A\bar{B}) = -aN(A)$ and $N(B) = (a^2 + r^2)N(A)$; for $q = -BA^{-1}$ these give $\operatorname{Re} q = a$ and $N(q) = a^2 + r^2$, so $q \in [a, r]$ and it is the required root.

Route two, by conjugating the remainder. Conjugate and multiply the remainder and reduce modulo $\Delta_{a,r}$; the result is exactly equation (5.5). If $\Delta_{a,r} \mid N_P$ both of its central coefficients vanish; if $A = 0$ positivity forces $B = 0$, and otherwise $q = -B\bar{A}/N(A)$ has precisely the scalar part and norm in equation (5.6).

Conversely, if $qA + B = 0$ at a point of the sphere then either $A = B = 0$, or $B = -qA$, in which case $N(B) = (a^2 + r^2)N(A)$ and $\operatorname{Re}(A\bar{B}) = -aN(A)$, so equation (5.5) vanishes and $\Delta_{a,r} \mid N_P$. $\square$

Theorem 5.3 (Complete zero-class criterion). *Let F be real closed and P nonzero of degree d . For each $a \in F$ and $r > 0$, P has a root on $[a, r]$ if and only if $\Delta_{a,r} \mid N_P$. When this divisibility holds, if $A = 0$ in equation (5.4) then $B = 0$ and the whole sphere consists of roots, and if $A \neq 0$ there is exactly one root there, $q = -BA^{-1}$. The central roots of P are exactly the roots of N_P in F . This is an effective algebraic algorithm relative to exact real-closed-field operations: form N_P , factor it over F , compute the two-coefficient remainders, and distinguish spherical from isolated roots.*

Proof. Lemma 5.2 gives every assertion about a fixed sphere. For central x the last claim is $N_P(x) = N(P(x))$ together with positivity of the norm. $\square$

Theorem 5.4 (Constructive fundamental theorem and factorization). *Over any real closed field F , every positive-degree one-sided $P \in \mathbb{H}_F[X]$ has a zero in $\mathbb{H}_F$ and factors as*

$$P = (X - q_1) \star (X - q_2) \star \cdots \star (X - q_d) a_d,$$

with the displayed order of factors significant. Its zero set meets at most d conjugacy classes, and in each noncentral class it is either a single point or the whole sphere. Its evaluation map $\mathbb{H}_F \rightarrow \mathbb{H}_F$ is surjective.

Proof. The nonconstant central polynomial N_P has a linear or an irreducible quadratic factor over F . A linear factor $X - x$ gives $0 = N_P(x) = N(P(x))$, hence $P(x) = 0$. Every monic irreducible quadratic over a real closed field is $X^2 - 2aX + a^2 + r^2$ with $r > 0$; apply theorem 5.3. Left division equation (5.2) reduces the degree, and induction gives the factorization. Every noncentral zero class contributes its distinct quadratic factor to N_P , and every central zero x makes all four $P_m(x)$ vanish, so $(X - x)^2 \mid N_P$; distinct classes supply relatively prime factors of degree two, and $\deg N_P = 2d$, so at most d classes occur. Applying the existence result to $P - b$ for each target b gives surjectivity. $\square$

This is a real-closed-field form of classical quaternionic zero theory — compare the slice-algebra treatment in [8] — proved with no appeal to topological degree on a compact sphere, which would need a separate justification over **No**. It supplies an actual finite root-recovery procedure from the central norm polynomial. It does not promise that factoring central polynomials over an arbitrary surreal presentation is computationally decidable.

Proposition 5.5 (Conservation of class multiplicity). *For a noncentral zero class $C = [a, r]$ put $m_C = \text{ord}_{\Delta_{a,r}} N_P$, and for a central root λ put $m_\lambda = \frac{1}{2} \text{ord}_{X-\lambda} N_P$, which is an integer because repeatedly dividing P by the central factor $X - \lambda$ removes a factor $(X - \lambda)^2$ from its normal. Then for $\deg P = d$,*

$$\sum_{C \text{ noncentral}} m_C + \sum_{\lambda \text{ central}} m_\lambda = d.$$

A spherical zero class has $m_C \geq 2$, so if s is the number of spherical classes and ℓ the number of remaining individual zeros, including central ones, then

$$2s + \ell \leq d. \tag{5.7}$$

Proof. Every irreducible factor of N_P corresponds to a zero by theorem 5.3; the degree of N_P is $2d$ and the indicated multiplicities count half its factor degrees. If the entire sphere vanishes, the remainder criterion gives $\Delta_{a,r} \mid P$, and normal multiplicativity then gives $\Delta_{a,r}^2 \mid N_P$. Every other zero class has multiplicity at least one, which implies equation (5.7). $\square$

Example 5.6 (Factor roots are not a list of evaluated roots). Let $P = (X - \mathbf{i}) \star (X - \mathbf{j}) = X^2 - X(\mathbf{i} + \mathbf{j}) + \mathbf{k}$. Then

$$P(\mathbf{i}) = 0, \quad P(\mathbf{j}) = 2\mathbf{k}, \quad N_P = (X^2 + 1)^2.$$

Modulo $X^2 + 1$ the remainder is $-X(\mathbf{i} + \mathbf{j}) + (\mathbf{k} - 1)$, whose unique root on the only possible class is $\mathbf{i}$; the whole zero set is the single point $\mathbf{i}$, an isolated root of class multiplicity two. The two written linear factors therefore do not supply two distinct pointwise zeros. In contrast $X^2 + 1$ vanishes on all of $\mathbb{S}(F)$. This example is a compact regression test that $\star$ is not pointwise multiplication.

Warning 5.7 (Two ways a noncommutative polynomial equation can fail to have a root). Theorem 5.4 cannot be restated for expressions with coefficients interspersed among the occurrences of the variable, and it cannot be restated for arbitrary noncommutative expressions. Two independent counterexamples:

- (1) $q + \mathbf{i}q\mathbf{i} + 1 = 0$ has no solution, since for $q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$ its left side is $1 + 2c\mathbf{j} + 2d\mathbf{k}$;
- (2) $\mathbf{i}q - q\mathbf{i} = 1$ has no solution, since its left side has zero scalar part.

The surquaternions are a noncommutative division algebra, not an algebraically closed field, and the one-sided, central-indeterminate convention is a mathematical hypothesis, not merely notation. Central polynomials behave differently again: there a noncentral root forces its entire conjugacy sphere to vanish.

6 Quaternion-valued Hahn series, valuation, and residue quaternions

6.1 Normal form and multiplication

Writing surreal normal forms with $t^\gamma = \omega^{-\gamma}$ turns the reverse-well-order convention into a well-order convention. Combining the four scalar Conway normal forms of a surquaternion and collecting coefficients gives the following.

Theorem 6.1 (Normal form, multiplication, localization). *Every surquaternion has a unique expression*

$$q = \sum_{\gamma \in S} t^\gamma h_\gamma, \quad h_\gamma \in \mathbb{H}_{\mathbb{R}} \setminus \{0\}, \quad (6.1)$$

with $S \subset \mathbf{No}$ a set-sized well-ordered support. Addition is coefficientwise and multiplication is ordered quaternion convolution,

$$(pq)_\eta = \sum_{\alpha+\beta=\eta} p_\alpha q_\beta, \quad (6.2)$$

each displayed coefficient sum being finite. Every set of surquaternions lies in one workspace D_Γ with Γ set-sized and divisible (theorem 1.1), and the coefficientwise identification

$$D_\Gamma = \mathbb{H}_{\mathbb{R}((t^\Gamma))} \simeq \mathbb{H}_{\mathbb{R}}((t^\Gamma)) \quad (6.3)$$

is an isomorphism of algebras with involution. The monomials t^γ are central.

Proof. A finite union of well-ordered subsets of a linearly ordered set is well-ordered, and uniqueness follows in each scalar coordinate. The ordinary Hahn sumset lemma makes the convolution locally

finite, and centrality of the monomials leaves the product of coefficients in its original order; finite coefficient rearrangement respecting that order proves associativity and distributivity. Localization is theorem 1.1, and real closedness of K_Γ for divisible Γ is the cited Hahn-field theorem [3, 6]. The four-coordinate construction proves the final isomorphism, which respects conjugation because the structure constants are integers. $\square$

Consequently equation (6.3) is a *Hahn series algebra with noncommutative coefficients, not a series construction with noncommuting exponents*. The order of h_α and k_β in equation (6.2) is essential; the value group is never noncentral, and all the noncommutativity sits in the $\mathbb{H}_\mathbb{R}$ -valued coefficients.

Definition 6.2 (Hahn-summable family). A *set-indexed* family $(q_j)_{j \in J}$ is *Hahn-summable*, equivalently *strongly summable*, if the union of its supports is well-ordered and each exponent belongs to the supports of only finitely many q_j . Its sum is defined coefficientwise, by adding those finitely many quaternion coefficients at each exponent. Every infinite sum below means this, unless an ordinary real or complex integral is explicitly named [5]. Ordinary convergence of infinitely many real coefficients is not a substitute, and this definition does not require convergence of partial sums in a topology. Such convergence may hold in a specified workspace, but is a separate assertion.

Theorem 6.3 (Exact valuation, leading coefficients, and residue). *For $q \neq 0$ put $w(q) = \min \text{supp}(q)$ and $\text{lc}(q) = h_{w(q)} \in \mathbb{H}_\mathbb{R}^\times$, with $w(0) = +\infty$. Then for nonzero p, q ,*

$$w(pq) = w(p) + w(q), \quad \text{lc}(pq) = \text{lc}(p) \text{lc}(q), \quad (6.4)$$

$$w(p + q) \geq \min\{w(p), w(q)\}, \quad w(\bar{q}) = w(q), \quad (6.5)$$

$$w(N(q)) = 2w(q), \quad w(|q|) = w(q), \quad (6.6)$$

and in coordinates $w(a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = \min\{w(a), w(b), w(c), w(d)\}$. The valuation ring and its maximal two-sided ideal are

$$\mathcal{O}_H = \{q : w(q) \geq 0\}, \quad \mathfrak{m}_H = \{q : w(q) > 0\},$$

consisting respectively of the quaternions with finite coordinates and with infinitesimal coordinates, where “finite” means bounded by some positive ordinary real. Coefficient extraction at exponent zero induces

$$\text{st} : \mathcal{O}_H \longrightarrow \mathbb{H}_\mathbb{R}, \quad \ker \text{st} = \mathfrak{m}_H, \quad \mathcal{O}_H / \mathfrak{m}_H \simeq \mathbb{H}_\mathbb{R}, \quad (6.7)$$

a surjective homomorphism of algebras with involution onto the noncommutative ordinary quaternion division algebra. Every finite q decomposes uniquely as $q = \text{st}(q) + \varepsilon$ with $\varepsilon \in \mathfrak{m}_H$.

Proof. At the smallest possible exponent in a product the coefficient is the product of the two leading coefficients, which is nonzero because $\mathbb{H}_\mathbb{R}$ is a division algebra; this proves equation (6.4), and the ultrametric inequality is coefficientwise. The leading coefficient of $N(q) = q\bar{q}$ is $N(\text{lc}(q)) > 0$, giving the first equality in equation (6.6), and taking its positive square root gives the second. A negative leading exponent is infinitely large, a positive one infinitesimal, and exponent zero carries a real quaternion coefficient followed by infinitesimal terms; on nonnegative supports only two zero exponents can contribute to the zero coefficient of a product, so st is a ring homomorphism with the indicated kernel and image. It commutes with conjugation because conjugation is coefficientwise. The coordinate formula is immediate from supports. $\square$

Thus $\mathcal{O}_H^\times$ consists exactly of the elements with nonzero standard part. The residue algebra is $\mathbb{H}_\mathbb{R}$, not $\mathbb{C}$ and not a commutative field: leading-coefficient noncancellation in equation (6.4) is a division-algebra argument, and the identity $\text{lc}(pq) = \text{lc}(p)\text{lc}(q)$ is *order-sensitive*. Valuation is invariant under inner conjugation, while its leading quaternion is conjugated by the leading quaternion of the conjugating element. The associated graded algebra in a workspace is the group-graded algebra with one copy of $\mathbb{H}_\mathbb{R}$ at each $\gamma \in \Gamma$, written $\mathbb{H}_\mathbb{R}[\Gamma]$ with central monomials.

Warning 6.4 (Valuation is not the modulus). The valuation is an additive order-of-magnitude measure, not the modulus: 1 and $2 + \mathbf{i}$ have the same valuation and different moduli. For matrices, only the inequality $w(AB) \geq w(A) + w(B)$ holds; the equality valid for individual nonzero quaternions must not be copied (see section 13.1).

6.2 Multiplicative normal form, the exact geometric inverse, and its certificate

Proposition 6.5 (Exact geometric inverse and error certificate). *Every nonzero q has a unique decomposition*

$$q = t^\gamma h(1 + \varepsilon), \quad \gamma = w(q), \quad h = \text{lc}(q) \in \mathbb{H}_\mathbb{R}^\times, \quad \varepsilon \in \mathfrak{m}_H, \quad (6.8)$$

and its inverse is the Hahn sum

$$q^{-1} = \left(\sum_{n \geq 0} (-\varepsilon)^n \right) h^{-1} t^{-\gamma}. \quad (6.9)$$

For nonzero $\varepsilon \in \mathfrak{m}_H$ and every integer $M \geq 0$,

$$w \left((1 + \varepsilon)^{-1} - \sum_{n=0}^M (-\varepsilon)^n \right) = (M + 1) w(\varepsilon). \quad (6.10)$$

The finite unit group $\mathcal{O}_H^\times$ is a split extension of $\mathbb{H}_\mathbb{R}^\times$ by $1 + \mathfrak{m}_H$, with the conjugation action displayed by $h(1 + \varepsilon) \cdot k(1 + \delta) = hk(1 + k^{-1}\varepsilon k)(1 + \delta)$.

Proof. Set $\varepsilon = h^{-1}t^{-\gamma}q - 1$; the leading term has been removed, so all remaining exponents are positive. Lemma 7.1 makes the geometric family Hahn-summable, and multiplication by $1 + \varepsilon$ telescopes coefficientwise to 1 on either side. The finite remainder is exactly $(-\varepsilon)^{M+1}(1 + \varepsilon)^{-1}$, whose final factor has valuation zero, so the valuation identities give equation (6.10). Reduction modulo $\mathfrak{m}_H$ supplies the split group map, and the displayed product exhibits the action. $\square$

The order in equation (6.9) is not cosmetic: in general h and ε do not commute. The multiplicative structure is a central monomial factor together with a semidirect product of principal units and ordinary nonzero quaternions, and the principal units are already nonabelian (section 8).

Example 6.6 (Two scales in one inverse). Let $t = \omega^{-1}$ and $q = 1 + t\mathbf{i} + t^2\mathbf{j}$. Then

$$q^{-1} = \frac{1 - t\mathbf{i} - t^2\mathbf{j}}{1 + t^2 + t^4},$$

and its formal geometric expansion agrees exactly with equation (6.9). The terms of the inverse are not obtained by independently inverting the four coordinate series. The absence of a $\mathbf{k}$ term here is the identity $\mathbf{i}\mathbf{j} + \mathbf{j}\mathbf{i} = 0$, not an approximation.

7 Summability certificates and the topology that must not be confused with them

7.1 The all-lengths positive-support lemma

Lemma 7.1 (Positive-support (Neumann) lemma, all lengths). *Let S be a well-ordered subset of the positive cone of an ordered abelian group. Then the generated additive monoid*

$$S^* = \{s_1 + \cdots + s_n : n \geq 0, s_m \in S\}$$

is well-ordered, and for each fixed exponent γ only finitely many finite ordered words $(s_1, \dots, s_n)$ over S have sum γ , counting all lengths at once. Consequently every noncommutative formal power series in finitely many variables with ordinary quaternion coefficients can be evaluated at finitely many elements of $\mathfrak{m}_H$, retaining the order of all factors.

Proof. A well-ordered alphabet is well-quasi-ordered. Higman’s finite-word lemma says that its finite words are well-quasi-ordered under subword embedding with entrywise domination. Such an embedding can only increase the sum of a word, and increases it strictly unless no entries are inserted and no entries increase, because all entries are positive. A strictly decreasing sequence of word sums would contradict the well-quasi-order property, so the sums are well-ordered; and distinct words with one fixed sum form an antichain, so there are only finitely many of them. (The usual choice principle is used to characterize a non-well-order by a decreasing sequence.)

For evaluation, take the finite union of the supports of the arguments. A contribution at exponent γ is specified by a word over this union, a choice of one of the finitely many variables at each position, and its coefficient term; there are only finitely many such contributions, and the possible exponents lie in S^* . $\square$

This all-lengths form, rather than a bound for each fixed word length, is the combinatorial input behind every infinitesimal power-series evaluation below; its use in surreal summation is discussed in [5], and the repository’s `Surreal/HahnSeries/Neumann.lean` proves the same all-lengths statement [12]. It is important that the proof works for arbitrary ordered abelian groups, not only rank-one or Archimedean ones.

Warning 7.2 (Positive valuation is a summability certificate, not cofinality). If $w(\varepsilon) = \delta > 0$ then $w(\varepsilon^n) = n\delta$, but the exponents $n\delta$ need not be cofinal in the value group: in the lexicographically ordered group $\mathbb{Z} \times \mathbb{Z}$ the sequence $n(0, 1)$ never exceeds $(1, 0)$. The geometric series is Hahn-summable all the same. Thus a remainder estimate $w(R_n) \geq n\delta$ is *not* automatically the assertion that $R_n \rightarrow 0$ in the valuation topology of the workspace, and a valid Hahn identity need not be the limit of its partial sums. Hahn summation avoids the ambiguity by specifying its coefficients directly.

7.2 The full surreal fine topology

The *fine topology* on $\mathbb{H}_{\mathbf{No}}$ has basic balls $B(q, r) = \{p : |p - q| < r\}$ for *every* positive surreal r , on the full surreal class. This is a ball-neighbourhood scheme; no collection whose elements are all open classes is required.

Theorem 7.3 (Set discreteness and the failure of sequential completion). *Every set $A \subset \mathbb{H}_{\mathbf{No}}$ is uniformly separated and closed in the fine topology. Every set-indexed fine-Cauchy net is eventually constant, and every convergent set-indexed net is eventually equal to its limit. Every continuous map from an ordinary connected real interval into $\mathbb{H}_{\mathbf{No}}$ is constant.*

Proof. The positive distances between distinct points of A form a set of positive surreals. Every set of positive surreals has a strictly positive lower bound — take a surreal larger than all their reciprocals and invert it; equivalently, the surreal cut property supplies a positive r with $2r$ smaller than all of them. Thus every ball of radius r contains at most one point of A . For $q \notin A$ the same argument applied to $\{|q - a| : a \in A\}$ supplies a ball at q disjoint from A . This proves closedness and discreteness, the empty and singleton cases being trivial.

A set-indexed net has a set as its image, and the Cauchy condition applied to a uniform separation radius forces all sufficiently late terms to coincide; if a net converges, apply the separation argument to its image together with the proposed limit. Finally a continuous interval map has a set-sized image, discrete in the induced topology, and a connected subset of a discrete space has at most one point. $\square$

This theorem is about the *full* surreal class and its fine topology. It is false as a blanket assertion about a set-sized Hahn field with its own *intrinsic* valuation topology: in $\mathbb{R}((t^{\mathbb{Q}}))$ one has $t^n \rightarrow 0$, because the integers are cofinal in $\mathbb{Q}$, and the geometric partial sums converge there to $(1 - t)^{-1}$. Inside **No**, however, every t^n exceeds t^ω , and neither assertion of convergence survives the finer neighbourhood test; the subspace topology that the entire workspace inherits from **No** is discrete, whereas its intrinsic topology is not.

Remark 7.4 (A workspace carries three different topologies). For $\Gamma \neq 0$, on K_Γ or D_Γ one must distinguish the *intrinsic* valuation topology, equivalently the topology of scalar modulus balls using radii in that workspace (the order topology on K_Γ); the discrete subspace topology inherited from the full fine topology; and the *standard-part* topology on the finite part, pulled back from $\mathbb{R}$ or $\mathbb{H}_{\mathbb{R}} \simeq \mathbb{R}^4$ by equation (6.7). The latter is not Hausdorff: 0 and a nonzero infinitesimal have the same standard part. For the stated equivalence of intrinsic topologies, choose $\delta > 0$ in Γ . Monomial radii bound valuation balls, and, for any workspace radius $r > 0$, a valuation ball with threshold greater than $w(r)$ lies inside the modulus ball of radius r ; conversely a modulus ball of radius $t^{\gamma+\delta}$ lies inside $\{x : w(x) > \gamma\}$.

The degenerate group $\Gamma = 0$ is an exception: $K_0 = \mathbb{R}$ and $D_0 = \mathbb{H}_{\mathbb{R}}$, the trivial valuation topology and the inherited full fine topology are discrete, whereas the modulus and standard-part topologies are the usual Hausdorff real or Euclidean ones. In the full fine topology no sequence of distinct rational approximations converges finely to $\sqrt{2}$, and the partial sums of $\sum \omega^{-n}$ do not converge finely to their Hahn sum. Coefficientwise analysis over ordinary domains uses its ordinary domain as a geometric base and does not pretend that its paths are fine-continuous surreal paths.

Example 7.5 (Zero fine differential does not imply constancy). On $\mathbb{H}_{\mathbf{No}}$ let $f(q) = 0$ when all coordinates of q are finite and $f(q) = 1$ otherwise. The finite-coordinate part is open, since a ball of radius 1 around one of its points stays finite; its complement is open, since at an infinite q the ball of radius $|q|/2$ stays infinite by the reverse triangle inequality. Hence f is locally constant and has zero fine differential at every point, but it is not globally constant.

The example is not a paradox: a mean-value argument would require a connectedness or integration theorem that this topology does not provide. For the rest of the article an infinite expression is marked by its construction — Hahn summation, ordinary real or complex analysis on coefficient functions, or a specified surreal operation — and the word “convergent” is never used as a substitute for one of those choices.

8 Infinitesimal Lie theory: exponential, logarithm, and rotation groups

8.1 The canonical neighbourhood of the identity

For $x, y \in \mathfrak{m}_H$ define

$$\text{Exp}_{\mathfrak{m}}(x) = \sum_{n \geq 0} \frac{x^n}{n!}, \quad \text{Log}_{\mathfrak{m}}(1 + y) = \sum_{n \geq 1} \frac{(-1)^{n+1} y^n}{n}, \quad (8.1)$$

both Hahn-summable by Lemma 7.1. Each individual series lies in the commutative subalgebra generated by its one argument together with the central scalars; the constant coefficient is handled as a single term, not as an ordinary convergent sum of constants, and the subscript emphasizes Hahn evaluation rather than a limit of partial sums.

Theorem 8.1 (Local exponential–logarithm equivalence). *The maps in equation (8.1) are mutually inverse bijections*

$$\text{Exp}_{\mathfrak{m}} : \mathfrak{m}_H \longleftrightarrow 1 + \mathfrak{m}_H : \text{Log}_{\mathfrak{m}}.$$

They commute with quaternion conjugation and with inner automorphisms, and for $x \neq 0$

$$w(\text{Exp}_{\mathfrak{m}}(x) - 1) = w(x), \quad \text{lc}(\text{Exp}_{\mathfrak{m}}(x) - 1) = \text{lc}(x).$$

If x and y commute then $\text{Exp}_{\mathfrak{m}}(x + y) = \text{Exp}_{\mathfrak{m}}(x) \text{Exp}_{\mathfrak{m}}(y)$. Their restrictions give a bijection

$$\text{Exp}_{\mathfrak{m}} : \mathfrak{m}_H \cap \text{Im } \mathbb{H}_{\text{No}} \longleftrightarrow U_1 : \text{Log}_{\mathfrak{m}}, \quad (8.2)$$

with U_1 as in equation (4.10).

Proof. The one-variable formal identities $\log(\exp T) = T$ and $\exp(\log(1 + T)) = 1 + T$ may be substituted at positive valuation: at each exponent only finitely many words contribute by Lemma 7.1, so formal regrouping becomes valid Hahn regrouping. Each identity involves powers of a single element, which generate a commutative subalgebra, so noncommutativity elsewhere is irrelevant. The leading coefficient of $\text{Exp}_{\mathfrak{m}}(x) - 1$ comes only from the linear term. Coefficientwise conjugation preserves these identities, and inner automorphisms preserve products and valuation; multiplying a Hahn-summable family by fixed factors h, h^{-1} preserves summability by the sumset lemma, although individual supports change.

If x is imaginary then $\overline{\text{Exp}_{\mathfrak{m}}(x)} = \text{Exp}_{\mathfrak{m}}(-x)$, so the modulus is 1. Conversely if $u \in U_1$ then $\bar{u} = u^{-1}$, hence $\overline{\text{Log}_{\mathfrak{m}} u} = \text{Log}_{\mathfrak{m}}(u^{-1}) = -\text{Log}_{\mathfrak{m}} u$ and the logarithm is imaginary. The commuting addition law follows from the formal binomial expansion and the same support regrouping. $\square$

This is a bijection, *not* an assertion that the additive group of imaginary infinitesimals is the group of infinitesimal rotations: the transported group law is noncommutative, and the correct law is BCH.

Proposition 8.2 (Baker–Campbell–Hausdorff at positive valuation). *For $x, y \in \mathfrak{m}_H$ the expression $\text{BCH}(x, y) = \text{Log}_{\mathfrak{m}}(\text{Exp}_{\mathfrak{m}}(x) \text{Exp}_{\mathfrak{m}}(y))$ is defined by Hahn summation, and*

$$\begin{aligned} \text{BCH}(x, y) = & x + y + \frac{1}{2}[x, y] + \frac{1}{12}[x, [x, y]] + \frac{1}{12}[y, [y, x]] \\ & - \frac{1}{24}[y, [x, [x, y]]] + R_5, \end{aligned} \quad (8.3)$$

where “degree” counts occurrences of x and y , not powers of a single numerical scale. If $w(x), w(y) \geq \delta > 0$ then the degree-four-and-higher remainder has valuation at least 4δ and $w(R_5) \geq 5\delta$. Writing $\alpha = w(x)$, $\beta = w(y)$ for nonzero x, y , the group commutator satisfies

$$\text{Exp}_{\mathfrak{m}} x \text{Exp}_{\mathfrak{m}} y \text{Exp}_{\mathfrak{m}}(-x) \text{Exp}_{\mathfrak{m}}(-y) = 1 + [x, y] + R, \quad w(R) \geq \alpha + \beta + \min\{\alpha, \beta\}. \quad (8.4)$$

Each exponential here is the infinitesimal one of Convention 2.1(a).

Proof. Use the formal exponential and logarithm in the free associative power-series algebra on two variables, filtered by total word degree. Their compositions have finitely many words at each degree, and expansion through degree four gives equation (8.3). At arguments of positive valuation every word is controlled by the monoid generated by their combined positive supports, and Lemma 7.1 makes evaluation and regrouping legitimate across all degrees; a common support certificate remains necessary for the whole infinite expression. Terms of degree at least n have valuation at least $n\delta$. In the group commutator the degree-two term is $xy - yx$, and every remaining nonzero word involves both variables and at least one additional occurrence, giving the stated bound. This derivation, rather than any analytic convergence theorem, is the one used. $\square$

For imaginary x, y one has $[x, y] = 2x \times y$. If their leading imaginary coefficients are not parallel, the commutator has valuation exactly $\alpha + \beta$ and is nonzero.

Example 8.3 (Rotations at unequal scales). For $x = t^\alpha \mathbf{i}$ and $y = t^\beta \mathbf{j}$ with $\alpha, \beta > 0$,

$$\text{BCH}(x, y) = t^\alpha \mathbf{i} + t^\beta \mathbf{j} + t^{\alpha+\beta} \mathbf{k} - \frac{1}{3} t^{2\alpha+\beta} \mathbf{j} - \frac{1}{3} t^{\alpha+2\beta} \mathbf{i} + \dots,$$

and the leading failure of commutativity appears at the sum of the two scales; there is no requirement that α and β be rationally comparable. For $\alpha = 1$, $\beta = 2$,

$$\text{Exp}_{\mathfrak{m}} x \text{Exp}_{\mathfrak{m}} y \text{Exp}_{\mathfrak{m}}(-x) \text{Exp}_{\mathfrak{m}}(-y) = 1 + 2t^3 \mathbf{k} + \text{terms of valuation at least } 4.$$

For a vector v the leading infinitesimal action is $\text{Exp}_{\mathfrak{m}}(x)v \text{Exp}_{\mathfrak{m}}(-x) = v + 2t^\alpha(\mathbf{i} \times v) + \dots$. Thus two infinitesimal rotations can agree with the identity to distinct orders and still produce a third, smaller rotational scale when commuted. The associated graded bracket is precisely twice the ordinary cross product, accompanied by addition of exponents. These bounds do not assume $n\delta$ cofinal and do not turn the Hahn identities into limits of a sequence in the fine topology (Warning 7.2).

Proposition 8.4 (Adjoint series at infinite arguments). For $x \in \mathfrak{m}_H$ and any fixed $q \in D_\Gamma$,

$$\text{Exp}_{\mathfrak{m}}(x) q \text{Exp}_{\mathfrak{m}}(-x) = \sum_{n \geq 0} \frac{\text{ad}_x^n(q)}{n!}, \quad \text{ad}_x(q) = [x, q], \quad (8.5)$$

and the series is Hahn-summable including when q is infinite.

Proof. The formal identity follows by collecting terms in $\sum_{r,s} x^r q(-x)^s / (r!s!)$. Its supports lie in $\text{supp}(q) + S^*$ with $S = \text{supp}(x)$, and the Hahn sumset lemma with Lemma 7.1 gives local finiteness. Noncommutativity is retained in the words $x^r q x^s$. $\square$

8.2 The filtered group of infinitesimal rotations

Theorem 8.5 (Logarithms of infinitesimal rotations). *The maps $\text{Exp}_{\mathfrak{m}}$ and $\text{Log}_{\mathfrak{m}}$ restrict to mutually inverse bijections between the imaginary infinitesimals $\text{Im}(\mathfrak{m}_H)$ and U_1 . On the left the group law is BCH, not addition. Every modulus-one u factors uniquely as $u = \text{st}(u)\omega_1$ with $\omega_1 \in U_1$, once this multiplication order is fixed; equivalently the sequence equation (4.10) splits, and every unit quaternion is finite, so this describes the whole modulus-one group.*

Proof. The bijection is equation (8.2). The factorization is $\omega_1 = \text{st}(u)^{-1}u$, which has standard part one and modulus one, and it is unique because st is a homomorphism on $\mathcal{O}_H^\times$. Every coordinate of a unit has absolute value at most 1, so units are finite. $\square$

Theorem 8.6 (Associated graded rotation algebra). *For $\gamma \in \Gamma_{>0}$ put $U_{\geq \gamma} = \{u \in U_1 : w(u-1) \geq \gamma\}$ and define $U_{> \gamma}$ similarly. Leading coefficients give*

$$U_{\geq \gamma}/U_{> \gamma} \simeq (\text{Im } \mathbb{H}_{\mathbb{R}}, +).$$

If $u = \text{Exp}_{\mathfrak{m}}(x)$ and $v = \text{Exp}_{\mathfrak{m}}(y)$ with x, y imaginary infinitesimals of valuations α, β and leading coefficients X, Y , then

$$uvu^{-1}v^{-1} = 1 + t^{\alpha+\beta}[X, Y] + \text{terms of valuation greater than } \alpha + \beta,$$

so its valuation above 1 is exactly $\alpha + \beta$ when $X \times Y \neq 0$, and the induced graded bracket is $[X, Y] = 2X \times Y$.

Proof. The product of $1 + a$ and $1 + b$ has leading correction $a + b$, since ab has larger valuation; the modulus-one condition forces the leading correction to be imaginary, and every such coefficient is realized by $\text{Exp}_{\mathfrak{m}}(t^\gamma X)$. Expanding the group commutator, its first nonzero formal term is $[x, y]$ by equation (8.4), and each subsequent term contains both arguments and at least one additional positive-valuation factor. This includes the possible cancellation when $[X, Y] = 0$. $\square$

9 Finite angles and polar coordinates at arbitrary scale

Let $\mathcal{O}_{\mathbf{No}}$ and $\mathfrak{m}_{\mathbf{No}}$ denote the finite and the infinitesimal surreal scalars. For a finite surreal $\theta = r + \eta$ with $r \in \mathbb{R}$ and $\eta \in \mathfrak{m}_{\mathbf{No}}$ define

$$\begin{aligned} \cos \theta &= \cos r \sum_{n \geq 0} \frac{(-1)^n \eta^{2n}}{(2n)!} - \sin r \sum_{n \geq 0} \frac{(-1)^n \eta^{2n+1}}{(2n+1)!}, \\ \sin \theta &= \sin r \sum_{n \geq 0} \frac{(-1)^n \eta^{2n}}{(2n)!} + \cos r \sum_{n \geq 0} \frac{(-1)^n \eta^{2n+1}}{(2n+1)!}. \end{aligned} \tag{9.1}$$

The ordinary real values are taken *first*, and only the infinitesimal correction is Hahn-summed; the real Taylor sums are not misrepresented as Hahn-summable families of constants. This is a restricted analytic extension, consistent with the established surreal analytic structure [4, 6] and with the repository's trigonometry report [14], whose scalar results we cite rather than reprove. Formal addition formulas and the ordinary real addition formulas prove their finite-surreal analogues. In the vocabulary of section 2, θ is a *finite angle* and η its *infinitesimal phase residue*.

Proposition 9.1 (Every direction has a finite angle; the finite phase theorem). *Fix I with $I^2 = -1$. The map $\theta \mapsto \cos \theta + I \sin \theta$ is a surjective homomorphism from $(\mathcal{O}_{\mathbf{No}}, +)$ onto the unit circle of $\mathbf{No} + \mathbf{No}I$ with kernel $2\pi\mathbb{Z}$. Equivalently: for $c, s \in \mathbf{No}$ with $c^2 + s^2 = 1$ there is a finite surreal θ , unique modulo $2\pi\mathbb{Z}$, with $c + Is = \cos \theta + I \sin \theta$, and if $s \geq 0$ there is a unique representative $\theta \in [0, \pi]$.*

Proof. The addition and norm identities are formal Taylor identities with a common positive support. The coordinates c, s are finite and their standard parts lie on the ordinary unit circle; choose an ordinary angle r of that standard part. Multiplication by $\cos r - I \sin r$ leaves a unit in $1 + \mathfrak{m}_H$ lying in the commutative slice $\mathbf{No} + \mathbf{No}I$, and its principal logarithm is $I\eta$ for a unique infinitesimal η by theorem 8.1; then $\theta = r + \eta$ works. If a phase equals 1, reducing to its standard part gives $r \in 2\pi\mathbb{Z}$, and injectivity of $\text{Log}_{\mathfrak{m}}$ forces $\eta = 0$; this is both the kernel computation and uniqueness modulo $2\pi\mathbb{Z}$. On the upper semicircle choose the ordinary representative in $[0, \pi]$; at its endpoints the sign of the infinitesimal sine correction selects the inward direction, so the corrected angle stays in the interval. The argument takes standard parts of the two slice coordinates, not of the symbol I , so it also works for a surreal imaginary unit. $\square$

Theorem 9.2 (Polar representation at arbitrary surreal radius). *Every nonzero surquaternion is expressible as*

$$q = \rho(\cos \theta + I \sin \theta), \quad \rho = |q| > 0, \quad \theta \in \mathcal{O}_{\mathbf{No}}, \quad I^2 = -1.$$

For noncentral q one may uniquely take $I = \text{Im } q / |\text{Im } q|$ and $0 < \theta < \pi$ with $\cos \theta = \text{Re } q / \rho$ and $\sin \theta = |\text{Im } q| / \rho$. Positive central elements use $\theta = 0$, negative ones $\theta = \pi$ with arbitrary I .

Proof. The normalized scalar and imaginary modulus lie on the unit circle with nonnegative second coordinate; apply Proposition 9.1. The sign of the sine selects the open upper semicircle and the unique representative in $(0, \pi)$, including the cases where the angle or its defect from π is infinitesimal, as is seen from equation (9.1). $\square$

An infinite modulus ρ therefore does not require an infinite angle: the surreal scale resides in the radius, and even an imaginary direction with infinitesimal coordinate components has a finite angle. Every unit surquaternion is $\cos \theta + I \sin \theta$ with $\theta \in [0, \pi]$, and formula equation (4.5) becomes rotation about I through the finite angle 2θ , as follows from the finite-angle double-angle identities — even for vectors of infinite or infinitesimal length.

10 Three exponentials: the tube, the global radial map, and their laws

The infinitesimal exponential $\text{Exp}_{\mathfrak{m}}$ was constructed in section 8. This section constructs the other two objects of Convention 2.1 and records exactly which law each satisfies. A global phase on all surreal scalars requires additional structure, and we make that structure explicit rather than treating $\sum q^n/n!$ as an everywhere-defined Hahn sum.

10.1 The finite-angle tube exponential

Let

$$\mathcal{T} = \{a + v \in \mathbb{H}_{\mathbf{No}} : a \in \mathbf{No}, v \in \text{Im } \mathbb{H}_{\mathbf{No}}, |v| \text{ finite}\}$$

and, using Gonshor's real exponential and the finite-angle trigonometry equation (9.1), define

$$\text{Exp}_{\mathcal{T}}(a + v) = \begin{cases} \exp_{\mathbf{No}}(a) \left(\cos r + \frac{v}{r} \sin r \right), & r = |v| > 0, \\ \exp_{\mathbf{No}}(a), & v = 0. \end{cases} \quad (10.1)$$

This is canonical within the chosen standard surreal exponential and restricted analytic structure. Its domain contains every finite quaternion and all central surreals, but *not* $\omega \mathbf{i}$.

Theorem 10.1 (Polar logarithms and arbitrary-scale roots). *The map $\text{Exp}_{\mathcal{T}}$ is conjugation-equivariant, satisfies $|\text{Exp}_{\mathcal{T}}(a + v)| = \exp_{\mathbf{No}}(a)$, obeys the addition law on commuting pairs in $\mathcal{T}$, and is surjective onto $\mathbb{H}_{\mathbf{No}}^{\times}$: if $q = \rho(\cos \theta + I \sin \theta)$ with $\rho = |q| > 0$ and $0 \leq \theta \leq \pi$ as in theorem 9.2, then*

$$\text{Exp}_{\mathcal{T}}(\log_{\mathbf{No}} \rho + I\theta) = q, \quad (10.2)$$

and for every ordinary integer $n \geq 1$ one n th root of q is

$$\rho^{1/n} (\cos(\theta/n) + I \sin(\theta/n)). \quad (10.3)$$

Proof. Inner conjugation fixes the scalar part and the modulus of the imaginary part and carries its direction to the conjugated direction. The modulus formula is $\cos^2 r + \sin^2 r = 1$. A commuting pair lies in a common slice by theorem 4.1; there equation (10.1) reads $\exp_{\mathbf{No}}(a)(\cos b + I \sin b)$ with a signed finite coordinate b , and the addition law is the finite sine and cosine addition formula. Every unit quaternion has the polar representation of theorem 9.2, which gives equation (10.2); applying the addition formula n times in the same slice gives equation (10.3). $\square$

There is no single-valued logarithm implicit here. On the negative scalar axis every $\log_{\mathbf{No}} \rho + \pi I$ with $I \in \mathbb{S}(\mathbf{No})$ exponentiates to $-\rho$, and even off that axis ordinary 2π periods give other logarithms. Surjectivity of a quaternion exponential need not entail a preferred logarithm.

On finite quaternions, equation (10.1) agrees with the four-real-variable Taylor extension of the ordinary quaternion exponential: at a nonzero ordinary imaginary radius this follows by analytic composition with the square root, and at radius zero the removable function $\sin r/r$ is evaluated by its even power series.

Warning 10.2 (The tube exponential is neither a power series nor a product). If $q = q_0 + \varepsilon$ with q_0 an ordinary quaternion and ε infinitesimal, then $\text{Exp}_{\mathcal{T}}(q)$ is *not* in general $\exp(q_0) \text{Exp}_{\mathbf{m}}(\varepsilon)$: the two summands need not commute. Nor is equation (10.1) obtained by Hahn-summing $\sum q^n/n!$ for every $q \in \mathcal{T}$, and the two failure modes are distinct: at a nonzero ordinary scalar all those terms have exponent zero, violating finite overlap in Definition 6.2; at the central $q = \omega \in \mathcal{T}$ their exponents are unbounded below. The latter failure also occurs for $q = \omega \mathbf{i}$, which lies outside the tube. Neither failure is repaired by citing ordinary real convergence.

10.2 The Ehrlich–Kaplan scalar normalization

Split a scalar normal form into its purely infinite and finite parts,

$$x = x_{\text{PI}} + \text{fin}(x), \quad \text{fin}(x) = r + \eta \in \mathcal{O}_{\mathbf{No}}.$$

The projection fin is *additive* but not multiplicative, and $\text{fin}(\omega) = 0$. Put

$$\text{Sin } x = \sin(\text{fin } x), \quad \text{Cos } x = \cos(\text{fin } x), \quad (10.4)$$

the canonical scalar phase functions associated with the initial integer part, the omnific integers

$$\mathbf{Oz} = \mathbf{No}_{\mathbf{PI}} + \mathbb{Z}.$$

Indeed $2\pi\mathbf{Oz} = \mathbf{No}_{\mathbf{PI}} + 2\pi\mathbb{Z}$, so reducing modulo this period group amounts exactly to discarding the purely infinite part and then reducing the finite angle; equivalently, for each $x \in \mathbf{No}$ one takes the unique $d \in \mathbf{Oz}$ with $x - 2\pi d \in [0, 2\pi)$ and evaluates the restricted real values there. This is the normalization of Ehrlich and Kaplan, Section 11 [6]. It gives

$$\sin \omega = 0, \quad \cos \omega = 1, \quad (10.5)$$

and on each complex slice a surjective exponential with period group $2\pi\mathbf{Oz}I$.

These are consequences of a *specified normalization*, not of ordered-field algebra alone, and the repository's trigonometry report [14] presents this character as one member of a family and explicitly does not claim it as new. We inherit that caveat: a different choice of global circular character would change the objects computed in theorems 10.5 and 11.2, and comparing such alternatives is left open in section 18. Global surreal trigonometry must not be dismissed as impossible; it must be named.

10.3 The global radial exponential

Definition 10.3 (Global radial surquaternion exponential). For $q = a + v$ with $r = |v|$, set

$$\text{Exp}_{\text{rad}}(q) = \begin{cases} \exp_{\mathbf{No}}(a)(\cos r + (v/r) \sin r), & r > 0, \\ \exp_{\mathbf{No}}(a), & r = 0. \end{cases} \quad (10.6)$$

The construction is *radial*: it uses the actual modulus r and the actual direction v/r of the imaginary part, and only then applies the chosen scalar phase to r . Discarding the purely infinite parts of the three vector coordinates first would define a different and in general non-equivariant operation. Both source manuscripts define exactly this map; one writes Exp_{EK} and the other Exp_{can} , and Convention 2.1(c) fixes the single name and symbol used here.

Theorem 10.4 (Properties of the global radial exponential). *The map equation (10.6) is a surjection $\mathbb{H}_{\mathbf{No}} \rightarrow \mathbb{H}_{\mathbf{No}}^{\times}$ satisfying*

$$\begin{aligned} |\text{Exp}_{\text{rad}}(q)| &= \exp_{\mathbf{No}}(\text{Re } q), & \text{Exp}_{\text{rad}}(-q) &= \text{Exp}_{\text{rad}}(q)^{-1}, & \text{Exp}_{\text{rad}}(\bar{q}) &= \overline{\text{Exp}_{\text{rad}}(q)}, \\ \text{Exp}_{\text{rad}}(h q h^{-1}) &= h \text{Exp}_{\text{rad}}(q) h^{-1} & (h \neq 0). \end{aligned}$$

It agrees with the finite-angle $\text{Exp}_{\mathcal{T}}$ on finite inputs and with $\text{Exp}_{\mathbf{m}}$ on $\mathbf{m}_H$, and it restricts on every slice $\mathbf{No} + \mathbf{No}I$ to the canonical surcomplex exponential, a surjective homomorphism with kernel $2\pi\mathbf{Oz}I$. It satisfies $\text{Exp}_{\text{rad}}(p + q) = \text{Exp}_{\text{rad}}(p) \text{Exp}_{\text{rad}}(q)$ whenever p and q commute. It is not an additive-to-multiplicative homomorphism on noncommuting arguments, and consequently its fibre over 1 must not be called a kernel. In particular, since $\omega/(2\pi) \in \mathbf{Oz}$,

$$\text{Exp}_{\text{rad}}(\omega \mathbf{i}) = 1. \quad (10.7)$$

Proof. The modulus, inverse and conjugation formulas follow from the scalar phase identities of section 10.2. Inner conjugation preserves a and r and carries the direction I to hIh^{-1} , proving equivariance. On a slice, writing $q = a + Ib$ turns equation (10.6) into $\exp_{\mathbf{No}}(a)(\cos b + I \sin b)$, using

oddness of Sin when $b < 0$; the slice is commutative, so the scalar phase homomorphism property and Proposition 9.1 give the slicewise homomorphism with the stated period group. Commuting noncentral quaternions lie in one slice by theorem 4.1, and central cases cause no change. Given $q \neq 0$, theorem 9.2 gives $\text{Exp}_{\text{rad}}(\log_{\mathbf{No}} |q| + I\theta) = q$, which is surjectivity. For infinitesimal $q = a + v$ the central scalar a commutes with v , and the even and odd powers of v are exactly the Hahn cosine and sine series at r , so expanding the commuting product gives the local series and hence agreement with Exp_{m} . Agreement with $\text{Exp}_{\mathcal{T}}$ on finite inputs holds because fin is the identity there. Failure of the global homomorphism law is witnessed by Example 10.6; the additive quaternion group is abelian while the multiplicative group is not, so no global homomorphism onto $\mathbb{H}_{\mathbf{No}}^{\times}$ is possible, and a noncommuting addition law is neither expected nor asserted. Finally $\text{fin}(\omega) = 0$ gives equation (10.5) and hence equation (10.7). $\square$

Equation (10.7) does not identify a divergent or nonsummable Taylor series with 1. It computes a value of a separately defined global operation, whose domain and law are those recorded in Convention 2.1(c).

Theorem 10.5 (Complete logarithm fibres). *For noncentral $q = \rho(\cos \theta + I \sin \theta)$ in the canonical polar form with $0 < \theta < \pi$,*

$$\text{Exp}_{\text{rad}}^{-1}(q) = \{\log_{\mathbf{No}} \rho + I(\theta + 2\pi z) : z \in \mathbf{Oz}\}. \quad (10.8)$$

For $\rho > 0$ the central fibres are

$$\text{Exp}_{\text{rad}}^{-1}(\rho) = \{\log_{\mathbf{No}} \rho + I(2\pi z) : I^2 = -1, z \in \mathbf{Oz}\}, \quad (10.9)$$

$$\text{Exp}_{\text{rad}}^{-1}(-\rho) = \{\log_{\mathbf{No}} \rho + I\pi(2z + 1) : I^2 = -1, z \in \mathbf{Oz}\}. \quad (10.10)$$

There is no preimage of zero.

Proof. The modulus formula determines the scalar part of every preimage. The image of a quaternion lies in its own slice; if the image is noncentral, its centralizer is that unique slice by theorem 4.1, so the preimage has the form $\log_{\mathbf{No}} \rho + Ib$ for a scalar b , and Proposition 9.1 with equation (10.4) makes its phase fibre exactly $b \in \theta + 2\pi\mathbf{Oz}$. When the image is central its imaginary part vanishes, so its radial phase is 1 or -1 and every axis is allowed, which gives equations (10.9) and (10.10), including the repetition of the scalar logarithm at $z = 0$ in the positive case. Moduli are strictly positive, excluding zero. $\square$

For positive and negative central values the fibres therefore sweep over *all* imaginary units I . These fibre descriptions do not settle the robustness questions for alternative surcomplex exponentials posed in [6].

Example 10.6 (Infinite radius is not coordinatewise phase reduction). With $q = \omega\mathbf{i} + \pi\mathbf{j}$,

$$r = \sqrt{\omega^2 + \pi^2} = \omega + \frac{\pi^2}{2\omega} + O(\omega^{-3}), \quad \text{so} \quad \text{Exp}_{\text{rad}}(q) = 1 + \frac{\pi^2}{2\omega}\mathbf{i} + O(\omega^{-2}),$$

whereas $\text{Exp}_{\text{rad}}(\omega\mathbf{i}) \text{Exp}_{\text{rad}}(\pi\mathbf{j}) = -1$. Here and below, order symbols for Hahn expansions bound the valuation of the remainder; they are not sequence limits. The example shows both that the direction and radius must be formed before the purely infinite phase is discarded, and that the homomorphism law genuinely fails at noncommuting arguments. Compare section 2.1: on either single slice the law does hold.

11 The differential of the radial exponential and its critical spheres

Differentiating a quaternionic function with respect to four scalar coordinates is not the same operation as applying a scalar-field derivation, and neither is the coefficientwise slice derivative. We use the following, which is Convention 2.2(c).

Definition 11.1 (Fine differential). An $\mathbf{No}$ -linear map L is the differential of f at q when for every surreal $\epsilon > 0$ there is a surreal $\delta > 0$ with

$$0 < |h| < \delta \implies |f(q+h) - f(q) - L(h)| < \epsilon |h|.$$

This is meaningful as a class scheme even though set-indexed fine limits are degenerate by theorem 7.3. It does not assert a real Banach-space structure, and a nonsingular fine differential is never used below as an inverse function theorem; the positive existence tool is theorem 15.1.

Theorem 11.2 (Radial derivative and critical spheres at infinite radius). *Let $q = a + rI$ with $r > 0$, and write a variation as $h = s + \alpha I + u$, where $s, \alpha \in \mathbf{No}$ and u is imaginary and perpendicular to I . Put $E = \exp_{\mathbf{No}}(a)$, $C = \cos r$, $S = \sin r$. Then*

$$d\text{Exp}_{\text{rad}}(q)[h] = E \left(s(C + IS) + \alpha(-S + IC) + \frac{S}{r} u \right), \quad (11.1)$$

and at a central point a the differential is $h \mapsto \exp_{\mathbf{No}}(a)h$. As a four-dimensional $\mathbf{No}$ -linear map,

$$\det d\text{Exp}_{\text{rad}}(a + rI) = \exp_{\mathbf{No}}(4a) \left(\frac{\sin r}{r} \right)^2 \quad (r > 0), \quad (11.2)$$

which is singular precisely for positive $r \in \pi\mathbf{Oz}$, and has rank exactly two there.

Proof. Fix q . For a variation $u' = \alpha I + u$ small relative to r , positive-square-root expansion gives

$$|rI + u'| = r + \alpha + O(|u'|^2/r), \quad \frac{rI + u'}{|rI + u'|} = I + \frac{u}{r} + O(|u'|^2/r^2),$$

Taylor identities on infinitesimal relative increments, justified by Lemma 7.1; alternatively the modulus expansion follows by rationalizing the difference of two square roots. The constants in their bounds may depend on the fixed surreal r . For infinitesimal scalar η the additive phase law gives $\cos(r + \eta) = C - S\eta + O(\eta^2)$ and $\sin(r + \eta) = S + C\eta + O(\eta^2)$ even when r is infinite, because fin is additive; likewise $\exp_{\mathbf{No}}(a + s) = E(1 + s + O(s^2))$. Combining the three expansions proves equation (11.1) with a remainder bounded by $M_q |h|^2$ on a sufficiently small ball for some positive surreal M_q , and shrinking the ball below $\epsilon/(M_q + 1)$ gives the differential in the sense of Definition 11.1. At $r = 0$ use the Hahn exponential at the small increment h together with the commuting central scalar a .

On the scalar–radial plane the matrix is $E \begin{pmatrix} C & -S \\ S & C \end{pmatrix}$, of determinant E^2 , and on the two transverse coordinates it is (ES/r) times the identity; this gives equation (11.2). Finally $\sin r = 0$ precisely when $\text{fin } r \in \pi\mathbb{Z}$, equivalently $r \in \pi\mathbf{Oz}$, and the scalar–radial block remains invertible, so a singular differential has rank exactly two. $\square$

Example 11.3 (A critical sphere of infinite radius). At $q = \omega I$ the normalized phase has $C = 1$, $S = 0$, so

$$d\text{Exp}_{\text{rad}}(\omega I)[s + \alpha I + u] = s + \alpha I.$$

Every transverse first-order variation vanishes while the scalar and radial ones survive. For a central formal small parameter η and $J \perp I$,

$$\text{Exp}_{\text{rad}}(\omega I + \eta J) = 1 + \frac{\eta^2}{2\omega}I + \frac{\eta^3}{2\omega^2}J + \text{terms of higher valuation}.$$

This is not obtained by exponentiating each imaginary coordinate independently; it is a radial critical-point phenomenon at an infinite scale.

The determinant formula extends the familiar finite-radius criticality of the quaternion exponential, but the allowed radii now include a proper class of infinite integer-part periods, the infinite radius ω among them. That this happens at all is a consequence of the chosen normalization of section 10.2, not of ordered-field algebra.

12 Derivations and the obstruction to an oscillatory calculus

12.1 Every algebraic derivation has a scalar and an inner part

Let $\delta : F \rightarrow F$ be a field derivation, meaning an additive map with $\delta(xy) = x\delta(y) + \delta(x)y$.

Theorem 12.1 (Derivation decomposition). *The formula*

$$\delta_{\mathbb{H}}(a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = \delta(a) + \delta(b)\mathbf{i} + \delta(c)\mathbf{j} + \delta(d)\mathbf{k} \quad (12.1)$$

defines the unique derivation of $\mathbb{H}_F$ extending δ and annihilating $\mathbf{i}, \mathbf{j}, \mathbf{k}$. It commutes with conjugation, and

$$\delta N(q) = 2 \operatorname{Re}(\delta_{\mathbb{H}}(q)\bar{q}), \quad \delta |q| = \frac{\operatorname{Re}(\delta_{\mathbb{H}}(q)\bar{q})}{|q|} \quad (q \neq 0). \quad (12.2)$$

Every ring derivation D of $\mathbb{H}_F$ has the unique form

$$D(q) = \delta_{\mathbb{H}}(q) + [w, q], \quad w \in \operatorname{Im} \mathbb{H}_F, \quad (12.3)$$

where δ is the restriction of D to the center, and the inner representative is explicitly

$$w = -\frac{1}{4}(E(\mathbf{i})\mathbf{i} + E(\mathbf{j})\mathbf{j} + E(\mathbf{k})\mathbf{k}), \quad E = D - \delta_{\mathbb{H}}. \quad (12.4)$$

Proof. Hamilton multiplication has integer structure constants, which proves the Leibniz rule for the coefficientwise extension and commutation with conjugation; differentiating $N(q) = q\bar{q}$ and then $|q|^2 = N(q)$ gives equation (12.2).

Differentiating $aq - qa = 0$ for central a shows that $D(a)$ commutes with every quaternion, so D restricts to a derivation of F ; subtracting the componentwise extension of that restriction leaves an F -linear derivation E . Two independent routes now identify E , and both were carried out in the source manuscripts.

Route one. For imaginary u , differentiate $u^2 = -N(u)$. Since E vanishes on F , this forces $E(u)$ to have zero scalar part and to be orthogonal to u (first for $u \neq 0$). Applying this to $u + v$ shows that E restricted to $V = \operatorname{Im} \mathbb{H}_F$ is skew-symmetric.

Route two. Differentiate $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1$ to see that $E(\mathbf{i}), E(\mathbf{j}), E(\mathbf{k})$ are imaginary and perpendicular to the corresponding basis vectors, then differentiate the anticommutation relations to see that the 3×3 matrix of E on V is skew-symmetric.

Either way, every skew-symmetric 3×3 matrix is uniquely $u \mapsto 2w \times u$ for a vector $w \in V$, by its three independent entries, and by Proposition 3.3 that map is $[w, \cdot]$; both maps vanish on F , which proves equations (12.3) and (12.4). Uniqueness follows because an imaginary element commuting with all of $\mathbb{H}_F$ is zero. $\square$

This is a concrete quaternionic form of the inner-derivation property of central simple algebras, and no infinite-dimensional algebra theorem is needed. For the componentwise extension one has

$$\delta_{\mathbb{H}}(q^{-1}) = -q^{-1}\delta_{\mathbb{H}}(q)q^{-1}, \quad \delta_{\mathbb{H}}(q^n) = \sum_{m=0}^{n-1} q^m(\delta_{\mathbb{H}}q)q^{n-1-m}. \quad (12.5)$$

The last formula becomes $nq^{n-1}\delta_{\mathbb{H}}q$ *only* when the relevant factors commute. The inner part in equation (12.3) gives an algebraic analogue of a change of quaternionic frame; its commutator cannot be absorbed into a scalar derivative of the coefficients.

12.2 The Berarducci–Mantova extension

Fix the Berarducci–Mantova derivation ∂ on **No** normalized by $\partial\omega = 1$. We use the established properties that it is strongly additive, commutes with the real surreal exponential, has constant field exactly $\mathbb{R}$, is surjective, and maps infinitesimals to infinitesimals [5, Theorems A and B]. These are external scalar results, not consequences of the quaternion construction. Let $\partial_{\mathbb{H}}$ denote the componentwise extension to $\mathbb{H}_{\mathbf{No}}$, with $\partial_{\mathbb{H}}\mathbf{i} = \partial_{\mathbb{H}}\mathbf{j} = \partial_{\mathbb{H}}\mathbf{k} = 0$.

Corollary 12.2 (Constants, surjectivity, antiderivatives). *$\ker \partial_{\mathbb{H}} = \mathbb{H}_{\mathbb{R}}$, so the constant algebra of $\partial_{\mathbb{H}}$ is exactly the ordinary quaternions; $\partial_{\mathbb{H}}$ is surjective, and every surquaternion has a coefficientwise antiderivative; and $\partial_{\mathbb{H}}$ respects Hahn-summable families coefficientwise.*

Proof. An element is annihilated exactly when each of its four coordinates lies in $\ker \partial = \mathbb{R}$. Choosing one scalar antiderivative for each of the four coordinates gives surjectivity, and strong additivity gives the last assertion. $\square$

Surjectivity of $\partial_{\mathbb{H}}$ is *not* a theorem that arbitrary nonlinear quaternionic differential equations have solutions, and a set-sized Hahn workspace need not be stable under ∂ : statements about a differential workspace require closure under that derivation or an explicit enlargement.

For an infinitesimal x , strong additivity of the *chosen Berarducci–Mantova* derivation justifies differentiating its Hahn exponential term by term, giving

$$\partial_{\mathbb{H}} \text{Exp}_{\mathbf{m}}(x) = \sum_{n \geq 1} \frac{1}{n!} \sum_{m=0}^{n-1} x^m \partial_{\mathbb{H}}(x) x^{n-1-m}, \quad (12.6)$$

whose right-hand family is support-controlled: it is built from words in the positive support of x with one distinguished factor from $\text{supp } \partial_{\mathbb{H}}(x)$, and Lemma 7.1 with the finite antidiagonals of sums of well-ordered sets justifies the regrouping. An arbitrary scalar derivation δ need not be strongly additive, so this passage does not follow merely from the algebraic extension theorem. For completeness, write $y = \partial_{\mathbb{H}}(x)$. The formal noncommutative exponential identity gives

$$\text{Exp}_{\mathbf{m}}(-x)\partial_{\mathbb{H}}\text{Exp}_{\mathbf{m}}(x) - y = -\frac{1}{2}[x, y] + \sum_{n \geq 2} \frac{(-1)^n}{(n+1)!} \text{ad}_x^n(y).$$

The same support argument justifies this evaluation. If $[x, y] \neq 0$, every term in the sum has valuation at least $w([x, y]) + w(x) > w([x, y])$, so cannot cancel the first term. Thus the simpler expression $\partial_{\mathbb{H}} \text{Exp}_{\mathfrak{m}}(x) = \text{Exp}_{\mathfrak{m}}(x) \partial_{\mathbb{H}}(x)$ holds exactly when $[x, \partial_{\mathbb{H}}(x)] = 0$ (the case $x = 0$ is immediate). This local assertion supplies no global commuting chain rule.

Proposition 12.3 (Angular velocity identity). *Let $u \in \mathbb{H}_F$ satisfy $N(u) = 1$ and set $\xi = \delta_{\mathbb{H}}(u)u^{-1}$. Then ξ is imaginary, and for every $x \in \mathbb{H}_F$*

$$\delta_{\mathbb{H}}(uxu^{-1}) = [\xi, uxu^{-1}] + u \delta_{\mathbb{H}}(x) u^{-1}. \quad (12.7)$$

For constant imaginary x the derivative of its rotated value is $2\xi \times (uxu^{-1})$.

Proof. Differentiate $u\bar{u} = 1$ to get $\xi + \bar{\xi} = 0$. The inverse rule is $\delta_{\mathbb{H}}(u^{-1}) = -u^{-1}\delta_{\mathbb{H}}(u)u^{-1}$; apply Leibniz to uxu^{-1} and collect terms. $\square$

This formula applies to surreal differential coefficients without pretending that the parameter is an ordinary real-time path into the fine topology. It provides an algebraic angular-velocity equation, not an existence theorem for trajectories.

12.3 No constant-frequency oscillation, and no commuting chain rule

Lemma 12.4 (Finite inputs have infinitesimal derivatives). *If $q \in \mathcal{O}_H$ then $\partial_{\mathbb{H}}q \in \mathfrak{m}_H$.*

Proof. Write $q = \text{st}(q) + \varepsilon$. The standard part lies in $\mathbb{H}_{\mathbb{R}}$ and is killed by $\partial_{\mathbb{H}}$. Each coordinate of ε is infinitesimal, and its derivative is infinitesimal by the smallness property of ∂ . The four-coordinate assertion follows. $\square$

Theorem 12.5 (No constant-frequency quaternionic oscillation). *Let $I \in \mathbb{S}(\mathbb{R})$ and $c \in \mathbb{R} \setminus \{0\}$. Neither equation*

$$\partial_{\mathbb{H}}y = cIy \quad \text{nor} \quad \partial_{\mathbb{H}}y = ycI \quad (12.8)$$

has a nonzero solution in $\mathbb{H}_{\mathbf{No}}$. In particular $\partial_{\mathbb{H}}y = \mathbf{i}y$ has only the zero solution, even inside a fixed surcomplex slice.

Proof. For the left equation,

$$\partial_{\mathbb{H}}N(y) = \overline{\partial_{\mathbb{H}}y}y + \bar{y}\partial_{\mathbb{H}}y = -\bar{y}cIy + \bar{y}cIy = 0,$$

and the right equation gives the same cancellation, using centrality of $N(y)$. Thus $N(y)$ lies in the constant field $\mathbb{R}$. If $y \neq 0$, this is a positive ordinary real number, so $|y| = \sqrt{N(y)}$ is a positive ordinary real and every coordinate of y is finite. By Lemma 12.4, $\partial_{\mathbb{H}}y$ has infinitesimal coordinates, so its modulus is infinitesimal or zero. But the right-hand side of either equation has modulus $|c| |y|$, a positive ordinary real. This is impossible. $\square$

Corollary 12.6 (No global commuting chain rule). *There is no map $E : \mathbb{H}_{\mathbf{No}} \rightarrow \mathbb{H}_{\mathbf{No}}^{\times}$ satisfying*

$$\partial_{\mathbb{H}}(E(q)) = E(q) \partial_{\mathbb{H}}q$$

for every q with $[q, \partial_{\mathbb{H}}q] = 0$, and the same conclusion holds with the factors on the right-hand side reversed. In particular the usual commuting exponential chain rule cannot hold at $q = \omega \mathbf{i}$.

Proof. At $q = \omega \mathbf{i}$ one has $\partial_{\mathbb{H}} q = \mathbf{i}$ and $[q, \partial_{\mathbb{H}} q] = 0$. A nonzero $E(q)$ would solve one of the equations forbidden by theorem 12.5. $\square$

For the global normalization of equation (10.6) the contradiction is visible as an exact identity:

$$\partial_{\mathbb{H}} \text{Exp}_{\text{rad}}(\omega \mathbf{i}) = \partial_{\mathbb{H}} 1 = 0, \quad \text{Exp}_{\text{rad}}(\omega \mathbf{i}) \partial_{\mathbb{H}}(\omega \mathbf{i}) = \mathbf{i}. \quad (12.9)$$

The normalization is still a valid global algebraic exponential, and by theorem 10.4 a genuine homomorphism on each commuting slice. It simply does not have this differential compatibility.

The proof of theorem 12.5 uses the constant field and the smallness property, not any assumption about the global phase. It therefore rules out an entire compatibility *requirement*, rather than exposing a defect of one phase choice: any proposed unified surquaternion calculus must respect this boundary.

Remark 12.7 (Relation to the companion results, and what is not being claimed). Three separations must be kept.

- (1) Theorem 12.5 is the componentwise quaternionic *extension* of “the missing oscillator” of [16], where the scalar and surcomplex statement — that $\partial^2 y + y = 0$ has only $y = 0$, and that $\sin \omega$ is not a Berarducci–Mantova differential element — is proved three independent ways. We cite that result and do not restate its proofs, and we do not present the quaternionic version as an independent discovery.
- (2) These conclusions do not conflict with the existence of real antiderivatives or with the Liouville-closed real differential-field properties of [5]: a constant imaginary frequency is not a scalar coefficient in that real differential field. Adjoining formal oscillations would require a larger differential algebra or a different derivation, and ordinary quaternion scalar extension by itself does not produce them.
- (3) Corollary 12.6 concerns the single derivation $\partial_{\mathbb{H}}$ and says nothing about the multiplicative behaviour of Exp_{rad} . It is therefore consistent both with the slice-wise homomorphism property of theorem 10.4 — hence with the commutative surcomplex homomorphism claim of [17] — and with the fine differential computed in theorem 11.2. See section 2.1.

Theorem 12.5 and Corollary 12.6 are proved here from the cited properties of the derivation. We make no claim that this formulation has never appeared in the literature; their role is to identify a concrete boundary.

13 Quaternionic linear algebra and multiscale spectral stability

The complex and real closed case of this material is already committed, in the companion report *Surcomplex Spectral Theory: Singular Values, Infinitesimal Rank, and Multiscale Stability* at `docs/surcomplex/spectral-theory/article.tex` [15]. That report proves Hermitian and normal diagonalization, positive square roots, the singular value decomposition, polar decomposition and the variational principles with attained extrema over a real closed field and its complexification, and over a Hahn workspace it proves gap-sensitive spectral projectors and subspace perturbation. We cite that report for those statements and print here only what noncommutativity changes.

What changes is the setting itself. On the *right* $\mathbb{H}_F$ -module $\mathbb{H}_F^n$ use

$$\langle x, y \rangle = \sum_{m=1}^n \bar{x}_m y_m, \quad |x| = \sqrt{\langle x, x \rangle} \in F_{\geq 0},$$

with matrices acting on the *left*, the adjoint $A^* = \bar{A}^T$ the conjugate transpose, and U unitary when $U^*U = I$. These conventions matter whenever eigenvalues or coefficients are moved across vectors: an eigenvalue must be written on the right, $Ax = x\lambda$, and it is a theorem, not a convention, that a Hermitian matrix has its eigenvalues in the *center* F . Orthogonalization uses finite sums and positive square roots, so quaternionic Gram–Schmidt is a finite algebraic procedure over a real closed field.

Theorem 13.1 (Hermitian spectral theorem over $\mathbb{H}_F$). *Let F be real closed and $A = A^* \in M_n(\mathbb{H}_F)$. There are a quaternionic unitary U and central scalars $\lambda_1, \dots, \lambda_n \in F$ with*

$$U^*AU = \text{diag}(\lambda_1, \dots, \lambda_n),$$

*and A is positive definite exactly when every $\lambda_m > 0$. Every positive definite Hermitian matrix has a unique positive definite Hermitian square root; every matrix admits a singular value decomposition with nonnegative central singular values; and every invertible square matrix has a polar decomposition $A = UP$ with U unitary and $P = (A^*A)^{1/2}$ positive definite Hermitian, uniquely.*

Proof. Write $A = Z + W\mathbf{j}$ with matrices over $C_F = F[\mathbf{i}]$ and extend equation (4.3) in blocks,

$$\rho(A) = \begin{pmatrix} Z & W \\ -\bar{W} & \bar{Z} \end{pmatrix},$$

which is Hermitian over the algebraically closed field C_F when A is. Its characteristic polynomial has a root $\lambda \in C_F$ with a nonzero eigenvector v ; since $v^*v > 0$ and $v^*\rho(A)v$ is fixed by conjugation, $\lambda = (v^*\rho(A)v)/(v^*v)$ lies in F . Two equivalent routes return from $\rho(A)$ to A : the identification $x + y\mathbf{j} \mapsto (x, -\bar{y})^T$ intertwines the action of $\rho(A)$ with that of A ; or, equivalently, every column $v \in C_F^{2n}$ is obtained by stacking the first columns of $\rho(x_m)$ for a unique $x \in \mathbb{H}_F^n$. Because λ is central, the matrix eigen-equation becomes $Ax = x\lambda$, a right eigenvector with the same central eigenvalue. Normalize x by the positive square root of its norm square. Its quaternionic orthogonal complement is A -invariant: if $\langle x, y \rangle = 0$ then $\langle x, Ay \rangle = \langle Ax, y \rangle = \lambda \langle x, y \rangle = 0$, using $A = A^*$ and centrality of λ . Quaternionic Gram–Schmidt produces an orthonormal basis of that complement and induction on n gives the diagonalization. The positivity criterion follows by evaluating the diagonal quadratic form, including on basis vectors.

For a positive definite matrix, replace each positive diagonal eigenvalue by its positive square root; uniqueness holds because a positive square root commutes with its square, preserves its eigenspaces, and on each eigenspace can only have the positive eigenvalue $\sqrt{\lambda}$. Diagonalizing A^*A gives singular values; for each nonzero singular value σ , normalize Av by σ and complete orthonormal bases of the remaining spaces, which proves the singular value decomposition. For invertible A , writing $A = (A(A^*A)^{-1/2})(A^*A)^{1/2}$ gives the polar factors. $\square$

No appeal to compactness of a surreal unit sphere occurs in this proof: it uses only finite linear algebra and algebraic closedness of the complexification of the center. Polar decomposition here is algebraic and does not assert compactness of $S^3(\mathbf{No})$. No infinite-dimensional Hilbert-space or proper-class spectral theorem is asserted anywhere.

13.1 Matrix modulus and its valuation

Define $\|A\|$ to be the largest singular value of A . The singular value decomposition proves $|Ax| \leq \|A\| |x|$ with equality for a suitable unit vector, and for a square $n \times n$ matrix

$$\max_{m,m'} |a_{mm'}| \leq \|A\| \leq n \max_{m,m'} |a_{mm'}|, \quad (13.1)$$

whence for a nonzero surquaternion matrix

$$w(\|A\|) = \min_{m,m'} w(a_{mm'}). \quad (13.2)$$

Matrix multiplication may cancel leading terms, so only $w(AB) \geq w(A) + w(B)$ is automatic; the equality of equation (6.4), valid for individual nonzero quaternions, must not be copied to matrices.

Warning 13.2 (Two determinant arguments that are unavailable here). The companion report [15] proves a determinantal singular-scale theorem, in which the k th determinantal valuation recovers the singular scales, and it uses a positive Cauchy–Binet expansion of the minors of a product. These proofs use the ordinary determinant of matrices over a commutative field; their minor identities cannot be transferred unchanged to $M_n(\mathbb{H}_F)$. Noncommutative determinants and the complex matrix realization provide other tools, but relating them to quaternionic singular scales would require a separate argument. No such determinant theorem is proved here; the multiscale results of this section use the min–max and projector arguments. That report’s explicit warning about Gram squaring applies here verbatim and is inherited, not restated.

13.2 Weyl’s estimate without compactness

Order the eigenvalues of a Hermitian matrix increasingly. Finite-dimensional spectral algebra gives the min–max formula

$$\lambda_k(A) = \min_{\dim W=k} \max_{x \in W, \|x\|=1} \langle x, Ax \rangle,$$

where the span of the first k eigenvectors realizes the upper bound, and any k -dimensional right subspace meets the span of eigenvectors $k, \dots, n$ nontrivially, proving the lower bound. The inner maximum exists by applying theorem 13.1 to the compressed Hermitian form on W . This notation therefore represents *attained extrema justified by linear algebra*, not by a general extreme-value theorem over F . Since $|\langle x, Ex \rangle| \leq \|E\|$ for unit x ,

$$|\lambda_k(A + E) - \lambda_k(A)| \leq \|E\| \quad (A = A^*, E = E^*). \quad (13.3)$$

If A has finite entries, its eigenvalues are finite by equation (13.1), and reducing a unitary diagonalization by standard part shows that their ordered standard parts are exactly the ordered eigenvalues of $\text{st}(A)$, with multiplicities.

Theorem 13.3 (Spectral-projector estimate at arbitrary surreal gap scale). *Let $A = A^* \in M_n(\mathbb{H}_F)$ with $n \geq 2$ have a simple eigenvalue λ , and let*

$$g = \min_{\mu \neq \lambda} |\mu - \lambda| > 0, \quad e = \|E\| < g/2, \quad E = E^*.$$

There is exactly one eigenvalue λ' of $A + E$ in $(\lambda - g/2, \lambda + g/2)$, counted with quaternionic multiplicity, and if P, P' are the associated rank-one orthogonal projections then

$$\|P' - P\| \leq \frac{e}{g - e}. \quad (13.4)$$

For surreal coefficients the sufficient condition $w(E) > w(g)$ gives

$$w(P' - P) \geq w(E) - w(g), \quad (13.5)$$

zero matrices being interpreted as having valuation $+\infty$.

Proof. Equation (13.3) places the matched ordered eigenvalue within e of λ and every other perturbed eigenvalue at least $g - e > g/2$ from λ , which proves uniqueness and simplicity. Let v' be a unit vector for λ' and write $v' = Pv' + y$ with $y = (I - P)v'$. Applying $I - P$ to $(A + E - \lambda')v' = 0$ gives $(A - \lambda')y = -(I - P)Ev'$. On $\text{im}(I - P)$ every spectral value of $A - \lambda'$ has absolute value at least $g - |\lambda' - \lambda| \geq g - e$, so theorem 13.1 gives $|y| \leq e/(g - e)$.

For two rank-one orthogonal projections, $\|P' - P\| = |(I - P)v'|$. To verify this over $\mathbb{H}_F$, multiply one generating vector on the right by a unit quaternion so that its inner product with the other is a nonnegative central scalar, and restrict to their two-dimensional span; the resulting real symmetric 2×2 matrix has eigenvalues $\pm \sqrt{1 - |\langle v, v' \rangle|^2}$, and the orthogonal complement contributes zero. This proves equation (13.4). If $w(E) > w(g)$ then e/g is infinitesimal, so $e < g/2$ and $w(g - e) = w(g)$; taking valuations and using equation (13.2) yields equation (13.5). $\square$

The estimate is useful exactly when the standard-part matrix has a multiple eigenvalue: an infinitesimal gap can still resolve its eigenvectors at a finer scale, and replacing g by an ordinary standard-part gap would discard precisely that information.

Example 13.4 (A sharp loss of gap scale). Let $t = \omega^{-1}$, $0 < \alpha < \beta$, and

$$A = \begin{pmatrix} 0 & t^\beta \mathbf{i} \\ -t^\beta \mathbf{i} & t^\alpha \end{pmatrix},$$

a perturbation of $\text{diag}(0, t^\alpha)$ of modulus t^β with gap t^α . Its eigenvalues are exactly

$$\lambda_\pm = \frac{t^\alpha \pm \sqrt{t^{2\alpha} + 4t^{2\beta}}}{2}, \quad (13.6)$$

and the positive-order binomial expansion gives

$$\begin{aligned} \lambda_- &= -t^{2\beta-\alpha} + t^{4\beta-3\alpha} + \text{terms of valuation at least } 6\beta - 5\alpha, \\ \lambda_+ &= t^\alpha + t^{2\beta-\alpha} - t^{4\beta-3\alpha} + \text{terms of valuation at least } 6\beta - 5\alpha. \end{aligned}$$

The lower spectral projector is $P_- = (\lambda_+ I - A)/(\lambda_+ - \lambda_-)$, whose off-diagonal entry is

$$-\frac{t^\beta \mathbf{i}}{\sqrt{t^{2\alpha} + 4t^{2\beta}}} = -t^{\beta-\alpha} \mathbf{i} + \text{terms of strictly higher valuation},$$

while its diagonal changes have valuation $2(\beta - \alpha)$. Thus $w(P_- - \text{diag}(1, 0)) = \beta - \alpha$, realizing the loss in equation (13.5) *exactly*, whereas the eigenvalue displacement occurs at the finer scale $2\beta - \alpha$. The standard-part matrix is degenerate, so an ordinary-gap version of the estimate would have discarded the information entirely.

Example 13.5 (An infinitesimal avoided crossing). For central $a, \delta \in F$ and $z \in \mathbb{H}_F$, the Hermitian matrix

$$A = \begin{pmatrix} a + \delta & z \\ \bar{z} & a - \delta \end{pmatrix}$$

satisfies $(A - aI)^2 = (\delta^2 + N(z))I$, so its eigenvalues are $a \pm \sqrt{\delta^2 + N(z)}$, the coincident case included. Taking $a = \omega$, $\delta = \omega^{-2}$ and $z = \omega^{-3} \mathbf{j}$ gives the exact gap $2\omega^{-2} \sqrt{1 + \omega^{-2}}$. The infinite common offset and the much smaller mixing scale cause no ambiguity in the algebraic spectrum. The phrase is borrowed; nothing is asserted here about atomic or molecular spectra.

Valuation error bounds of this kind are exact symbolic data. Truncating a Neumann inverse after degree M certifies equation (6.10), and a spectral perturbation can carry both $w(E)$ and the gap valuation $w(g)$ to certify equation (13.5). These are stronger than a floating-point residual when the important effect lives below every ordinary real scale. They do not by themselves certify a limit in any topology, least of all one whose value group has larger scales than all multiples of the current step (Warning 7.2). Substituting a small positive *real* t in Example 13.4 should be treated as an illustration, not as the definition of the surreal problem.

14 Support-controlled analysis: slice series, coherent families, Cauchy and Fueter

A satisfactory analytic theory must relate the many complex slices without confusing the four operators of Convention 2.2: the coefficientwise slice derivative ∂_{sl} , the field derivation $\partial_{\mathbb{H}}$, the fine differential of Definition 11.1, and the Fueter operator $\mathcal{D}$. They have different domains and different defining properties. This section constructs a controlled extension of classical slice analysis [8]; every infinite expression uses Hahn supports or ordinary coefficient analysis, and no assertion about arbitrary class functions on $\mathbb{H}_{\mathbf{No}}$ is intended.

14.1 Admissible series on an infinitesimal disk

Consider right-coefficient series

$$f(x) = \sum_{n \geq 0} x^n a_n. \quad (14.1)$$

If every $a_n \in \mathbb{H}_{\mathbb{R}}$ and $x \in \mathfrak{m}_H$, this is Hahn-summable regardless of the ordinary real radius of convergence of the coefficients: infinitesimal support, not a real analytic estimate, does the work. More generally one may take $a_n \in \mathcal{O}_D$ under the *common-support hypothesis* that all $\text{supp}(a_n)$ lie in one well-ordered set $S \subseteq \Gamma_{\geq 0}$.

Proposition 14.1 (Uniform-support evaluation). *Under the common-support hypothesis, equation (14.1) is Hahn-summable at every $x \in \mathfrak{m}_H$ in a common workspace, and so is its formal derivative $\partial_{\text{sl}} f = \sum_{n \geq 1} n x^{n-1} a_n$. Products and formal differentiation are compatible with Hahn regrouping whenever the same common-support conditions hold for the participating series.*

Proof. Let $T = \text{supp}(x)$, which is positive. Every summand is supported in $S + nT$. The monoid T^* is well-ordered by Lemma 7.1, and $S + T^*$ is well-ordered. For a fixed exponent there are only finitely many pairs consisting of an exponent of S and one of T^* , and only finitely many words representing the latter across all lengths, so only finitely many summands contribute. Shifting the index by one proves the derivative assertion, and the same word count proves the regroupings and the formal product rule. $\square$

Warning 14.2 (The slice derivative is not the differential). The derivative in Proposition 14.1 is the *slice or formal* derivative ∂_{sl} of Convention 2.2(b). It is not a claim that $df_x(h) = h \partial_{\text{sl}} f(x)$ for arbitrary noncommuting h : even the polynomial x^2 has full algebraic differential $h \mapsto xh + hx$ rather than $h \mapsto 2hx$ in all directions, as Proposition 15.2 makes exact.

Warning 14.3 (The common-support hypothesis is substantive, and so is the coefficient class). Two independent failures must be recorded.

- (1) Knowing $w(a_n) \geq 0$ individually is *not* a substitute for a common S . For a counterexample with integral coefficients, take $\Gamma = \mathbb{Q} \times \mathbb{Q}$ in lexicographic order, realized inside **No** by $(a, b) \mapsto a\omega + b$, and put $x = t^{(0,1)}$ and $a_n = t^{(1, -n^2)}$ for $n \geq 1$. Every a_n has positive valuation, but $x^n a_n = t^{(1, n - n^2)}$ has a strictly decreasing sequence of exponents for $n \geq 1$, so the family is not Hahn-summable.

A separate failure, allowing nonintegral coefficients, illustrates dependence on the workspace. In $\Gamma = \mathbb{Q}$, the series $\sum_{n \geq 0} t^{-n^2} x^n$ is not Hahn-summable at any nonzero argument: if $w(x) = \delta \in \mathbb{Q}$ then the leading exponents $-n^2 + n\delta$ are unbounded below. Enlarging the workspace can change this: at the surreal argument $x = t^\omega$ those leading exponents increase and the monomial family is Hahn-summable. A failure in a fixed value group is not automatically a failure at every surreal scale.

- (2) Ordinary quaternion coefficients and Hahn-valued coefficients are different classes and must not be interchanged. A nonzero germ with coefficients in $\mathbb{H}_{\mathbb{R}}$ has a first nonzero coefficient a_m , and then $f(q) = q^m(a_m + \varepsilon)$ with $\varepsilon \in \mathfrak{m}_H$ for every nonzero infinitesimal q , so f does not vanish; Hahn-valued coefficients allow genuinely displaced infinitesimal zeros, such as the zero of $q - t^\gamma \mathbf{i}$.

A central translation and scaling replace x by $(q - a)/r$ with $a, r \in \mathbf{No}$, $r \neq 0$, giving controlled infinitesimal disks even around an infinite central center. It does not supply arbitrary expansions about noncentral centers: the order of increments and center terms then matters and must be retained in noncommutative words.

Proposition 14.4 (Slice representation, Cauchy–Riemann, twisted product). *For an admissible series f , central infinitesimal x, y and any $I, J \in \mathbb{S}(\mathbf{No})$ in the workspace,*

$$f(x + yJ) = \frac{1}{2}(1 - JI)f(x + yI) + \frac{1}{2}(1 + JI)f(x - yI), \quad (14.2)$$

and on a fixed slice $(\partial_x + I\partial_y)f(x + yI) = 0$ coefficientwise. The formal convolution product obeys the twisted evaluation law equation (5.1) whenever the value used for conjugation is invertible, and gives zero when that value is zero.

Proof. For central x, y the powers of $x + yI$ have the form $A_n(x, y) + IB_n(x, y)$ with real integer-coefficient polynomials A_n, B_n independent of I ; summing gives $f(x + yI) = A + IB$, and solving the two equations with I and $-I$ produces equation (14.2). The Cauchy–Riemann and derivative identities hold for every monomial and then by common-support regrouping. For the product formula, repeat the proof of Lemma 5.1; inner conjugation leaves the valuation and modulus of the infinitesimal argument unchanged, and a support certificate follows from the fixed factors $f(q)^{\pm 1}$ together with the supports of the original evaluated terms. $\square$

14.2 Common-domain coefficient families: two formalizations

For analytic continuation and contours one needs a *single* ordinary domain shared by all coefficients. The two source manuscripts set this up in two genuinely different ways, and both are kept here because they support different theorems.

Formalization A: Hahn stems with an auxiliary central unit. Let $\Omega \subset \mathbb{C}$ be an ordinary connected conjugation-invariant open set meeting the real axis, and introduce an *auxiliary central*

complex unit ι , distinct from every quaternion unit. An ordinary *stem coefficient* is a holomorphic map

$$F_\gamma : \Omega \rightarrow \mathbb{H}_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \quad \text{with} \quad F_\gamma(\bar{z}) = \overline{F_\gamma(z)}^\iota,$$

where the overline changes ι to $-\iota$ and leaves ordinary quaternion coefficients unchanged; the target is a complexified quaternion algebra, not the division algebra $\mathbb{H}_{\mathbf{No}}$. A *common-domain Hahn stem* is

$$F(z) = \sum_{\gamma \in S} t^\gamma F_\gamma(z), \quad (14.3)$$

with S well-ordered and the same Ω for every coefficient. At a finite $z = z_0 + \eta$ with $z_0 = \text{st}(z) \in \Omega$, evaluate by

$$F(z) = \sum_{\gamma \in S} \sum_{n \geq 0} t^\gamma \frac{F_\gamma^{(n)}(z_0)}{n!} \eta^n, \quad (14.4)$$

the factors η being central in this complexification; the support proof of Proposition 14.1 makes the double family Hahn-summable, requiring no common bound on the magnitudes of all ordinary derivatives, since finiteness at each exponent replaces an analytic sum over γ . Writing $F(a + \iota b) = A(a, b) + \iota B(a, b)$ with $A, B \in \mathbb{H}_{\mathbf{No}}$, the induced *slice function* is

$$f(a + Ib) = A(a, b) + IB(a, b), \quad I \in \mathbb{S}(\mathbf{No}), \quad (14.5)$$

for finite a, b with $\text{st}(a + \iota b) \in \Omega$. The parities $A(a, -b) = A(a, b)$ and $B(a, -b) = -B(a, b)$ make this independent of the representation by $(-I, -b)$, and at $b = 0$ the vanishing of B makes it independent of I altogether. Values may be infinite, since S may contain negative exponents, even though the argument domain here is a finite standard-part thickening. The Cauchy–Riemann identities for the stem are

$$A_a = B_b, \quad A_b = -B_a, \quad (14.6)$$

so on a fixed slice $\frac{1}{2}(\partial_a + I\partial_b)f = 0$. These variable derivatives mean ordinary coefficient derivatives followed by the same Taylor evaluation; they are not $\partial_{\mathbb{H}}$ acting on the surreal coefficients as numbers.

Formalization B: coherent families over an ordinary circular quaternionic domain. Fix instead an ordinary circular quaternionic domain $\Omega \subseteq \mathbb{H}_{\mathbb{R}}$ and a well-ordered $S \subseteq \Gamma$, and consider

$$f = \sum_{\gamma \in S} t^\gamma f_\gamma, \quad f_\gamma : \Omega \rightarrow \mathbb{H}_{\mathbb{R}} \text{ ordinary stem-induced slice-regular functions}, \quad (14.7)$$

with Ω common to all coefficients; slice regularity is in the stem-function sense on a conjugation-invariant complex domain, $f_\gamma(x + Iy) = A_\gamma(x, y) + IB_\gamma(x, y)$ with quaternion-valued components satisfying equation (14.6) and the even/odd parity that makes the definition independent of the sign of I . For finite x, y with standard part in that domain, evaluate A_γ, B_γ by their ordinary Taylor germs at the standard point, substitute the positive Hahn corrections, and then sum over γ ; the support certificate is $S + T^*$ with T the union of the supports of the finitely many scalar increments, and the extra factor I is a fixed finite quaternion contributing only its own well-ordered support. Uniqueness of ordinary Taylor germs ensures that shrinking a neighbourhood does not change the value.

Both formalizations are closed under the slice product induced by multiplying stem functions, and under coefficientwise differentiation, with support sets transformed by finite sums and sumsets. In neither is the slice product the pointwise product of quaternion values. A family of analytic germs with no common ordinary domain is a *different space* and supports neither contour theorem below.

Theorem 14.5 (Representation formula and coherent identity principle). *For every slice function of the form equation (14.5) and all $I, J \in \mathbb{S}(\mathbf{No})$,*

$$f(a + Ib) = \frac{1}{2}[(1 - JI)f(a + Ib) + (1 + JI)f(a - Ib)]. \quad (14.8)$$

If a common-domain Hahn slice function vanishes on an ordinary nonempty real interval in Ω , then it vanishes throughout its induced domain.

Proof. Substituting $A + IB$ and $A - IB$ on the right of equation (14.8) gives $A + JB$, no coefficient having been moved across I or J . At an ordinary real point s the function is $\sum_{\gamma} t^{\gamma} F_{\gamma}(s)$ with ordinary quaternion coefficients; if it is zero then uniqueness of Hahn coefficients makes every $F_{\gamma}(s)$ zero. Vanishing on a real interval together with the ordinary complex identity theorem, applied to its four holomorphic coordinates, makes each F_{γ} zero on connected Ω , and evaluation gives zero everywhere. $\square$

This identity principle is deliberately coefficientwise. It does not claim that an arbitrary fine-differentiable class function with many zeros is identically zero; Example 7.5 already shows the danger of such a claim.

14.3 Products, reciprocals, and zero spheres

If two stems have values $A + \iota B$ and $C + \iota D_1$, their product induces

$$(f \star g)(a + Ib) = (AC - BD_1) + I(AD_1 + BC), \quad (14.9)$$

the slice product, which need not equal $(A + IB)(C + ID_1)$. Supports of the product lie in the sum of two well-ordered sets with finite coefficient decompositions, so the class is closed under it. At a point where $f(q) \neq 0$ the same twist as in equation (5.1) holds:

$$(f \star g)(q) = f(q) g(f(q)^{-1} q f(q)), \quad (14.10)$$

since conjugating $q = a + Ib$ replaces I by $I' = f(q)^{-1} I f(q)$ while leaving a, b fixed, and then $f(q)(C + I'D_1) = f(q)C + I f(q)D_1$ expands to equation (14.9); if $f(q) = 0$, substitution gives $(f \star g)(q) = 0$.

Quaternion-conjugating the coefficients of a stem defines f^c , and its normal $f^s = f \star f^c = f^c \star f$ has central-complex stem values,

$$(A + \iota B)(\bar{A} + \iota \bar{B}) = N(A) - N(B) + 2\iota \operatorname{Re}(A\bar{B}),$$

so f^s is *slice-preserving*: at $a + Ib$ its value lies in $\mathbf{No} + \mathbf{No}I$, not necessarily in the center $\mathbf{No}$. Wherever the normal is nonzero and a reciprocal stem is admitted locally, the slice inverse evaluates as

$$f^{-\star}(q) = f^s(q)^{-1} f^c(q), \quad (14.11)$$

and the displayed order is part of the formula. Local admission requires support control: on an ordinary neighbourhood where a leading central coefficient of the normal has no zeros, factor out that coefficient and monomial and invert the positive-order remainder by Proposition 6.5. A pointwise nonzero normal is by itself no proof that one global common-domain reciprocal family exists without shrinking or changing domains.

On one conjugacy sphere the value is $A + IB$: if $B = 0$, either the entire sphere vanishes or none of it does; if $B \neq 0$, at most one point vanishes, the only candidate being $I = -AB^{-1}$, which must also satisfy $I^2 = -1$. This is the analytic counterpart of Lemma 5.2, and it gives the local shape of the zero set, not a finiteness theorem for the total number of zeros of every analytic function.

14.4 Two coefficientwise Cauchy formulas

Both manuscripts prove a coefficientwise Cauchy formula, and the two results are genuinely different — different kernels, different generality in the argument, different hypotheses — so both are printed. In each, the integral of a common-support Hahn family along an ordinary contour is *defined* by integrating each ordinary quaternion coefficient function; this is a definition of a functional, not a fine-topological limit of quaternion Riemann sums.

Theorem 14.6 (Common-domain Cauchy formula at arbitrary slice direction). *Fix an ordinary $I_0 \in \mathbb{S}(\mathbb{R})$ and a bounded ordinary conjugation-invariant domain U whose closure lies in Ω and whose positively oriented boundary is piecewise smooth; write U_{I_0} for its copy in $\mathbb{R} + \mathbb{R}I_0$. Put*

$$K(s, q) = -(q^2 - 2\operatorname{Re}(s)q + |s|^2)^{-1}(q - \bar{s}). \quad (14.12)$$

Then for a common-domain Hahn slice function as in Formalization A and any finite q in the standard-part thickening of the circularization of U ,

$$f(q) = \frac{1}{2\pi} \int_{\partial U_{I_0}}^H K(s, q) (-I_0 ds) f(s), \quad (14.13)$$

the superscript H denoting the coefficientwise construction just specified.

Proof. If the standard-part conjugacy sphere of q comes from a point of U , the quadratic denominator in equation (14.12) is nonzero for $s \in \partial U_{I_0}$, and the symmetry of U ensures that both ordinary slice representatives lie in the interior rather than on the boundary.

First take ordinary $q \in U_{I_0}$. The commuting factors in its slice make $K(s, q) = (s - q)^{-1}$. Each coefficient of f restricted to that slice is an ordinary left-complex-holomorphic quaternion-valued function; splitting its values into two complex components gives the ordinary Cauchy formula with multiplication on the displayed sides, and all ordinary derivatives of that formula hold as well, so Taylor evaluation proves it at infinitesimally displaced points of this fixed slice.

For a fixed general finite q , expand the rational kernel about $q_0 = \operatorname{st}(q)$. On the ordinary compact boundary its zeroth-order denominator has no zero, so its reciprocal admits a Neumann expansion in the finitely many infinitesimal coordinates of $q - q_0$ by Proposition 6.5. All kernel coefficients are ordinary smooth functions on the contour, and all supports lie in the positive monoid generated by those infinitesimal supports; multiplying by the common support S of $f(s)$ produces a well-ordered support with finite overlap, uniformly along the contour as a support certificate. Thus the stated coefficientwise integral exists and the coefficient identities may be integrated.

Finally $K(s, q)$ is a slice function of q , its quadratic factor having real coefficients and its numerator being affine with the constant on the right. Apply equation (14.8) to both sides of equation (14.13) and use the fixed-slice result at $a \pm I_0 b$ to obtain the equality at $a + Jb$ for every $J \in \mathbb{S}(\mathbf{No})$. $\square$

Theorem 14.7 (Coefficientwise Cauchy formula on a fixed slice). *Fix an ordinary imaginary unit I and an ordinary positively oriented piecewise smooth Jordan contour C whose interior and boundary lie in $\Omega \cap (\mathbb{R} + \mathbb{R}I)$, so that each f_γ of equation (14.7) is holomorphic on a neighbourhood of that closed interior. If q_0 is an ordinary interior point and $q = q_0 + \eta$ with infinitesimal $\eta \in K_\Gamma + IK_\Gamma$, then*

$$f(q) = \frac{1}{2\pi} \int_C (s - q)^{-1} ds_I f(s), \quad ds_I = -I ds, \quad (14.14)$$

the integral and its infinitesimal kernel expansion being defined coefficientwise.

Proof. On the slice, split an ordinary quaternion-valued holomorphic function into two complex-valued holomorphic components with a fixed perpendicular unit on the right. The ordinary complex Cauchy formula proves equation (14.14) for each f_γ at q_0 and gives all its derivatives there. Expand $(s - q_0 - \eta)^{-1} = \sum_{n \geq 0} \eta^n (s - q_0)^{-n-1}$; the contour stays a positive ordinary distance from q_0 , so each coefficient integral is an ordinary well-defined integral equal to the corresponding Taylor coefficient. Supports of the total family lie in $S + \text{supp}(\eta)^*$ and are locally finite by Lemma 7.1. Thus coefficientwise integration reconstructs exactly the defined value $f(q)$. $\square$

Only ordinary contours in coefficient space have been used. The compactness appealed to concerns the real contour, not a surreal unit sphere, and neither formula is an integral along a fine-continuous surquaternion path. The kernel equation (14.12) is the classical slice Cauchy kernel [8]; the additional work here supplies the Hahn support certificate and clarifies what the integral means over surreal coefficients. No claim is made that testing an arbitrary surquaternion function at ordinary points determines it: such uniqueness is a property of the specified coherent coefficient representation, not of all functions on $\mathbb{H}_{\mathbf{No}}$.

Example 14.8 (Coefficient germs without a common domain). A common domain cannot be removed from the hypotheses merely by saying that every coefficient is an analytic germ. For $\eta > 0$,

$$F(z) = \sum_{r \geq 1} \frac{t^{r\eta}}{1 - rz} \quad (14.15)$$

has a convergent germ for each coefficient, but its coefficient poles $1/r$ approach 0: there is no ordinary disk about 0 on which all of them are holomorphic, and no common small circle on which the holomorphic-inside-contour argument applies. Formal evaluation on a suitable infinitesimal disk can still make sense — that is a different function class, precisely the distinction emphasized in the repository's document catalogue [13].

14.5 Fueter regularity is a different condition

For quaternion coordinates $q = a + x_1\mathbf{i} + x_2\mathbf{j} + x_3\mathbf{k}$ define the left Fueter operator and the four-dimensional Laplacian by

$$\mathcal{D} = \partial_a + \mathbf{i}\partial_{x_1} + \mathbf{j}\partial_{x_2} + \mathbf{k}\partial_{x_3}, \quad \Delta_4 = \partial_a^2 + \partial_{x_1}^2 + \partial_{x_2}^2 + \partial_{x_3}^2,$$

with coefficients multiplying on the left. Again the partial derivatives are *variable* operators on ordinary coefficient functions and their Taylor extensions, never $\partial_{\mathbb{H}}$. A function is left Fueter regular when $\mathcal{D}f = 0$. Let $r = \sqrt{x_1^2 + x_2^2 + x_3^2}$ and $I = (x_1\mathbf{i} + x_2\mathbf{j} + x_3\mathbf{k})/r$ away from the real axis. For an axial slice function $f = A(a, r) + IB(a, r)$ satisfying equation (14.6), direct coordinate differentiation gives

$$\mathcal{D}f = \left(A_a - B_r - \frac{2B}{r}\right) + I(B_a + A_r) = -\frac{2B}{r}, \quad (14.16)$$

$$\Delta_4 f = \frac{2A_r}{r} + I\left(\frac{2B_r}{r} - \frac{2B}{r^2}\right), \quad (14.17)$$

so a slice-regular function need not be Fueter-regular: the coordinate function $f(q) = q$ is slice regular with $\mathcal{D}f = -2$. The two theories are not interchangeable.

Theorem 14.9 (Fueter transfer: polynomials and coherent families). *For every right-coefficient slice polynomial P over a real closed field,*

$$\mathcal{D}\Delta_4 P = 0, \quad (14.18)$$

and the same identity holds coefficientwise for a common-domain Hahn slice function constructed from holomorphic stems, in either formalization of section 14.2:

$$\mathcal{D}\Delta_4 f = 0. \quad (14.19)$$

The formula holds away from the real axis and extends across that axis wherever the stem is defined on an ordinary neighbourhood there, the extension being coefficientwise analytic.

Proof. Set $C = 2A_r/r$ and $E = 2B_r/r - 2B/r^2$, so that $\Delta_4 f = C + IE$ by equation (14.17), using that A and B are harmonic in (a, r) . The Fueter operator of an arbitrary axial function is

$$\mathcal{D}(C + IE) = (C_a - E_r - 2E/r) + I(E_a + C_r),$$

and using $A_a = B_r$, $B_a = -A_r$ and their differentiated forms,

$$\begin{aligned} C_a - E_r - \frac{2E}{r} &= \frac{2B_{rr}}{r} - \left(\frac{2B_{rr}}{r} - \frac{4B_r}{r^2} + \frac{4B}{r^3} \right) - \left(\frac{4B_r}{r^2} - \frac{4B}{r^3} \right) = 0, \\ E_a + C_r &= \frac{2B_{ar}}{r} - \frac{2B_a}{r^2} + \frac{2A_{rr}}{r} - \frac{2A_r}{r^2} = 0. \end{aligned}$$

Hence $\mathcal{D}\Delta_4 f = 0$ for $r > 0$. The parity of a stem makes A even and B odd in r , so the apparent divisions in equation (14.17) have removable analytic singularities at $r = 0$; for example $B_r/r - B/r^2$ cancels the contribution of the linear term in B . For a polynomial the identity extends over the real axis as a polynomial identity, and, read as an identity in the four scalar coordinates, it consequently holds over every real closed field, which is equation (14.18). For an ordinary slice-regular coefficient the components are real analytic, so the identity extends across real points by analyticity. Differentiation of coefficient functions does not enlarge their exponent support, and Taylor evaluation is Hahn-summable, so all identities extend coefficientwise to the whole family. $\square$

This is the classical Fueter transformation placed in a specified surreal setting, not a new claim about the classical theorem [8]. Its derivation also makes the convention visible: right-handed regularity would require the corresponding right-handed operators and product order. For concrete checks,

$$\mathcal{D}q = -2, \quad \Delta_4(q^2) = -4, \quad \Delta_4(q^3) = -4(3a + x_1\mathbf{i} + x_2\mathbf{j} + x_3\mathbf{k}),$$

and applying $\mathcal{D}$ to either right-hand side gives zero; multiplying such a slice polynomial by t^γ or taking a common-support Hahn combination preserves the coefficientwise theorem. The recorded suites check $\mathcal{D}\Delta_4(q^n) = 0$ for $n = 0, \dots, 7$, which is a genuinely quaternionic signature and not a surcomplex fact.

14.6 What has and has not been constructed

We now have formal infinitesimal functions, two classes of common-domain coefficient families, their slice products, a coherent identity principle, two Cauchy functionals and a Fueter transformation. The full surreal quaternion class does not thereby become an ordinary Banach-space manifold. We have not asserted an unrestricted maximum principle, a global residue theorem on arbitrary

surreal paths, or the existence of a Taylor expansion for every fine-differentiable function; the identity principle is deliberately coefficientwise, and coefficientwise integration gives no uniqueness for arbitrary functions on $\mathbb{H}_{\mathbf{No}}$. For an application requiring such a statement, its function class and its integration or support notion must be specified first.

15 Support-controlled implicit lifting and square-root conditioning

This section gives the positive existence mechanism for solving perturbed quaternionic equations. Its essential hypothesis is nonsingularity of the *full real-coordinate Jacobian*; a nonzero derivative on a single complex slice is not enough. For a scalar vector $x = (x_1, \dots, x_d)$ put $w(x) = \min_m w(x_m)$ with $w(0) = +\infty$, and write $\mathcal{O}_K, \mathfrak{m}_K$ for the valuation ring and its maximal ideal in K_Γ .

Theorem 15.1 (Support-controlled implicit lifting). *Let $P \in \mathcal{O}_K[X_1, \dots, X_d]^d$ and let $x_0 \in \mathbb{R}^d$ satisfy*

$$\text{st } P(x_0) = 0, \quad J_0 = \left(\frac{\partial \text{st } P_m}{\partial X_{m'}}(x_0) \right)_{m,m'} \in \text{GL}_d(\mathbb{R}).$$

Then there is exactly one $h \in \mathfrak{m}_K^d$ with $P(x_0 + h) = 0$. Let $S \subset \Gamma_{>0}$ be the union of the supports of the positive-order parts of the finitely many coefficients of $P(x_0 + X)$. Then

$$\text{supp}(h_m) \subseteq S^* \setminus \{0\} \quad (1 \leq m \leq d),$$

and if $P(x_0) \neq 0$ then $w(h) = w(P(x_0))$. Uniqueness holds among all infinitesimal vectors. No countable convergence or cofinality assumption on Γ is required.

Proof. Write the finitely many coefficient perturbations as $e_1, \dots, e_{m_0} \in \mathfrak{m}_K$ and replace them by independent central variables $Y_1, \dots, Y_{m_0}$. This gives a polynomial system $\mathcal{P}(X, Y)$ over $\mathbb{R}$ with $\mathcal{P}(0, 0) = 0$ and X -Jacobian J_0 at $(0, 0)$.

There is a unique formal solution $H(Y) \in (Y)\mathbb{R}[[Y]]^d$, by the following complete recursion. If all homogeneous terms of H below degree n have been determined, the degree- n part of $\mathcal{P}(H(Y), Y)$ is $J_0 H_n + R_n$, where R_n depends only on the already determined terms; set $H_n = -J_0^{-1} R_n$. Each degree involves only finitely many monomials, which determines a formal solution and proves its formal uniqueness.

Now evaluate $Y_m = e_m$. By Lemma 7.1 the family of all evaluated monomials is Hahn-summable: its supports lie in S^* and each exponent has only finitely many contributing words. Thus $h = H(e_1, \dots, e_{m_0})$ exists as a Hahn vector, has the asserted support, and satisfies $P(x_0 + h) = 0$ because finite polynomial operations respect this evaluation.

To prove uniqueness among all infinitesimal vectors, not merely among those obtained by this substitution, let $h, h' \in \mathfrak{m}_K^d$. Polynomial divided differences give

$$P(x_0 + h) - P(x_0 + h') = (J_0 + E)(h - h'), \quad E \in M_d(\mathfrak{m}_K),$$

and the determinant of $J_0 + E$ has nonzero real constant coefficient, so the matrix is invertible and two roots coincide.

Finally put $\varrho = P(x_0)$. The equation for h is $\varrho + J_0 h + E_1 h + R_{\geq 2}(h) = 0$ with E_1 of positive order and all coefficients of $R_{\geq 2}$ integral. If $h \neq 0$, both error terms have valuation strictly greater than $w(h)$; an invertible real matrix preserves vector valuation, so cancellation at the least exponent forces $w(\varrho) = w(h)$. If $\varrho = 0$, uniqueness gives $h = 0$. $\square$

This is a support-explicit form of nonsingular Hensel lifting. Its extra usefulness here is that quaternion multiplication is polynomial in four scalar coordinates, so any finite system of quaternion equations — including expressions with coefficients inserted on both sides of the unknowns — becomes such a scalar system, and theorem 15.1 applies as soon as its real-coordinate Jacobian is nonsingular. This does *not* give a root theorem for every noncommutative polynomial expression: Warning 5.7 still applies, and the interspersed equation $q + \mathbf{i}q\mathbf{i} + 1 = 0$ has no solution at all.

15.1 The exact Sylvester operator for squaring

Let $q_0 = a + rI \neq 0$ with $r > 0$ and $I^2 = -1$. Every increment has the unique orthogonal decomposition

$$h = h_{\parallel} + h_{\perp}, \quad h_{\parallel} = \frac{h - IhI}{2} \in F + FI, \quad h_{\perp} = \frac{h + IhI}{2},$$

where $h_{\perp}$ is imaginary and perpendicular to I .

Proposition 15.2 (Square-root linearization and condition numbers). *The differential of $q \mapsto q^2$ at q_0 is*

$$L_{q_0}(h) = q_0h + hq_0 = 2q_0h_{\parallel} + 2a h_{\perp}, \quad (15.1)$$

with determinant, as an F -linear map on F^4 ,

$$\det L_{q_0} = 16a^2 N(q_0).$$

For $a \neq 0$ its inverse is

$$L_{q_0}^{-1}(u) = (2q_0)^{-1}u_{\parallel} + (2a)^{-1}u_{\perp}, \quad (15.2)$$

its singular values are $2|q_0|$ twice and $2|a|$ twice, and if $a = 0$ its kernel is exactly the two-dimensional perpendicular plane.

Proof. The parallel component commutes with I and the perpendicular component anticommutes with it, which proves equation (15.1) by expansion. On the parallel two-plane, left multiplication by $2q_0$ is a similarity of scale $2|q_0|$ and determinant $4N(q_0)$; on the perpendicular two-plane the map is multiplication by $2a$. The remaining assertions follow. $\square$

A purely imaginary square root can therefore have a perfectly nonzero *slice* derivative $2q_0$ while its four-dimensional differential has a two-dimensional kernel. This is the linear-algebraic explanation for the spheres of square roots of a negative central scalar in Proposition 4.2, and the reason one cannot import a complex simple-root test without modification. Note that the “condition number” language here is an exact valuation-theoretic statement about $L_{q_0}^{-1}$, not a numerical-analysis heuristic.

Example 15.3 (A noncommuting displaced square root). Let t be a positive infinitesimal monomial and $q_0 = 1 + \mathbf{i}$. There is a unique root close to q_0 of $q^2 = q_0^2 + t\mathbf{j}$, whose first terms are

$$q = 1 + \mathbf{i} + \frac{t}{2}\mathbf{j} + \frac{t^2}{16}(1 - \mathbf{i}) - \frac{t^3}{32}\mathbf{j} + O(t^4), \quad (15.3)$$

where $O(t^4)$ denotes a Hahn remainder of valuation at least $4w(t)$, not an ordinary limiting process. Substituting $q = q_0 + \sum_{n \geq 1} t^n h_n$ gives the exact recursion

$$L_{q_0}(h_1) = \mathbf{j}, \quad L_{q_0}(h_n) = - \sum_{m=1}^{n-1} h_m h_{n-m} \quad (n \geq 2),$$

with support in $\mathbb{N}w(t)$; theorem 15.1 justifies the whole series rather than only the displayed truncation.

15.2 A quantified near-singular lifting region

The next result keeps track of the small perpendicular singular value, and is useful even when q_0 itself has no nonsingular real reduction.

Theorem 15.4 (A sufficient conditioned square-root threshold). *Let $a \in \mathfrak{m}_K$ be nonzero, put $\kappa = w(a) > 0$ and $q_0 = a + \mathbf{i}$. For $e \in D_\Gamma$ with $\sigma = w(e) > 2\kappa$ there is exactly one solution of*

$$q^2 = q_0^2 + e$$

with $w(q - q_0) > \kappa$, and writing $h = q - q_0$,

$$w(h) \geq \sigma - \kappa, \quad w(h - L_{q_0}^{-1}e) \geq 2\sigma - 3\kappa. \quad (15.4)$$

The condition $\sigma > 2\kappa$ is asserted as sufficient, not as an optimal threshold for every direction of e .

Proof. Write $h = az$. The equation $L_{q_0}h + h^2 = e$ becomes $z + aL_{q_0}^{-1}(z^2) = a^{-1}L_{q_0}^{-1}e$. The real-coordinate matrix of $aL_{q_0}^{-1}$ has integral entries: its parallel block is multiplication by $a/(2q_0)$ and its perpendicular block is $1/2$; moreover every entry of $L_{q_0}^{-1}$ has valuation at least $-\kappa$ by Proposition 15.2. The right-hand side therefore has valuation at least $\sigma - 2\kappa > 0$. The system in z has real reduction vanishing at 0 with Jacobian the identity, so theorem 15.1 gives a unique infinitesimal z with $w(z) \geq \sigma - 2\kappa$, which is the first estimate for h . The exact identity $h - L_{q_0}^{-1}e = -L_{q_0}^{-1}(h^2)$ and multiplicativity of the valuation give the second. The condition $w(h) > \kappa$ is precisely $z \in \mathfrak{m}_K$, which also proves the stated uniqueness region. $\square$

The theorem identifies the relevant loss as that of the *inverse linearized operator*. The determinant alone records a product of four scales and obscures which perturbation directions are ill-conditioned: parallel errors and perpendicular errors can behave very differently.

16 Exact computation and formalization routes

A surquaternion is represented by four scalar objects together with their common field or workspace certificate, and addition, conjugation, multiplication, norm square and inversion reduce to finitely many scalar operations. The difficulty is never Hamilton multiplication; it is what the scalar backend can prove about surreal coefficients and infinite supports. No finite-string format can encode every member of the proper class **No**, and even within a set-sized Hahn field an arbitrary support or coefficient sequence need not be computable.

16.1 A representation hierarchy

Scalar model	Exact quaternion operations	Additional requirements
Rational functions in central monomials	Four-coordinate arithmetic, conjugation, inverse, normal polynomials, Cayley and stereographic charts	Ordered monomial group and scalar equality
Real algebraic extensions of that field	Modulus, square roots, finite real-closed algebra, spectral equations	Certified algebraic roots and order
Effective Hahn series with support certificates	Neumann inverse, infinitesimal exponential and logarithm, finite coefficient extraction, implicit lifting to a stated degree	Decidable support queries and finite-overlap witnesses
Common-domain analytic coefficient families	Slice product, Taylor evaluation, coefficientwise Cauchy and Fueter operators	Domain certificates and exact coefficient analytic operations
Unrestricted symbolic surreal expressions	Finite denotation of some further values	No blanket decidability or evaluation guarantee

The division formula $q^{-1} = \bar{q}/N(q)$ should normally be the first implementation path: it needs one central inverse, not a noncommutative linear-system solver. Positive roots and moduli should be delegated to the scalar real-closed model, and an implementation must not replace the F -valued modulus by a machine floating-point norm. A rational Hahn model is not closed under every square root; ramification or algebraic extension may be required.

For Hahn objects, maintain well-ordered support data, a leading exponent and an ordinary quaternion leading coefficient; inversion first removes the leading monomial and coefficient and then uses the positive-order geometric series equation (6.9). For an exact infinite object, a support certificate and a coefficient rule are part of its meaning, and a finite truncation should state the valuation of its omitted remainder, as in equation (6.10). Such truncations need not provide all coefficients below an arbitrary cutoff in finitely many steps: an infinite support can have infinitely many exponents below that cutoff.

A series evaluator must return a value together with a summability certificate, or reject the request as unsupported in that representation. It must not silently reinterpret $\sum(\omega \mathbf{i})^n/n!$ as a global exponential (Warning 10.2). Likewise a polynomial implementation must keep the $\star$ product separate from pointwise function multiplication; Example 5.6 is a compact regression test for exactly that distinction. Polynomial root computation has a precise algebraic route — form N_P , factor it over the real closed scalar field into linear and irreducible quadratic factors, and compute the two-term remainder modulo each quadratic, which decides point versus sphere and recovers an isolated root as $-BA^{-1}$ — but its effectiveness depends on the scalar field’s factorization facilities. For implicit lifting, the finite formal recursion of theorem 15.1 is implementable up to a chosen formal degree; if supports have several independent scales, a formal-degree cutoff and a valuation cutoff are different specifications, and a correct program records both the computed expression and the support or valuation information proving what has been omitted.

16.2 Two proposed formalization layerings

The historical repository snapshot cited by the source manuscripts uses generic algebra and Hahn-series infrastructure rather than a completed surreal construction, and its README explicitly says that its generic Lean prerequisites do *not* yet construct the surreal field or the normal-form

bridge [12]. We neither identify those prerequisites with a formalization of this article nor infer a theorem from a successful build reported elsewhere. Mathlib already provides a generic quaternion algebra with conjugation, norm square and division-ring instances under ordered-field hypotheses [11]; an initial formalization should instantiate and extend those objects rather than duplicate their multiplication proofs, and in particular a commutative complexification abstraction alone does not encode the noncommuting quaternion units. The following are *proposed layers, not claims that corresponding modules exist or compile*. The current theorem-by-theorem status is maintained separately in `docs/FORMALIZATION.md`; at this review all 48 indexed statements of this report remain pending. Both source manuscripts proposed a layering, and they differ in granularity and in where they place the surreal bridge; both orderings are recorded.

Layering I, five layers, bridge last. *Layer A: finite quaternion algebra.* Work over a generic ordered or real closed coefficient field, formalizing the norm, inverse, centralizer, conjugacy criterion, central-polynomial division and the normal-polynomial zero algorithm *before* introducing surreal numbers, with all statements quantified over a set-sized type. *Layer B: geometry and matrices.* Add the F -valued modulus, unit quaternions, the rotation action, Hermitian diagonalization, Weyl's inequality and the projector estimate, avoiding an ordinary real-normed-space abstraction that would incorrectly force the modulus into $\mathbb{R}$. *Layer C: quaternion Hahn series.* Prove the four-coordinate equivalence $\mathbb{H}_{\mathbb{R}((t^\Gamma))} \simeq \mathbb{H}_{\mathbb{R}}((t^\Gamma))$ for a type-sized ordered abelian group Γ , reusing the repository's support, regrouping and standard-part theorems; the new obligations are ordered noncommutative coefficient convolution, leading-coefficient noncancellation, the norm valuation identity, and the infinitesimal exponential/logarithm substitution lemmas. *Layer D: the actual surreal bridge.* Supply a separately justified surreal construction and normal-form interpretation with an explicit universe or class discipline, and only then specialize the generic results and prove localization; generic Hahn lemmas alone do not establish this bridge, and no set-sized type should be advertised as containing every surreal number at the same foundational level. *Layer E: analytic structures and their incompatibilities.* Formalize common-domain stems and the coefficientwise integral using an ordinary complex analytic library, and independently formalize the chosen surreal derivation and the properties used in theorem 12.5; the no-oscillation argument is short once those foundations exist. A global trigonometric normalization belongs in another explicitly named structure, and one must not give it an unjustified instance of a chain-rule axiom.

Layering II, six layers, with lifting and coherent analysis separated.

Proposed layer	Main statements and dependencies
Quaternion algebra	Conjugacy, slices, one-sided polynomial reduction, finite spectral theory; generic ordered or real closed scalar field.
Quaternion Hahn series	Coefficientwise equivalence, multiplicative valuation, residue algebra; four-coordinate transport of support lemmas.
Infinitesimal Lie theory	Exponential, logarithm, BCH and unit-group filtration; Neumann finiteness for all word lengths.
Support-controlled lifting	Formal multivariable recursion and its evaluation; real-coordinate Jacobian certificate.
Coherent slice analysis	Common ordinary domains, coefficientwise contour operations, representation and Fueter identities.
Full surreal bridge	Class or universe discipline, normal forms, scalar exponential, finite part, and the chosen global phase.

The first five layers of Layering II can be meaningfully proved in set-sized generic workspaces; the last is not obtained by declaring a small type whose elements are all surreal numbers. Both layerings agree on the essential point: the surreal bridge comes *after* the generic Hahn work, and the ordered magnitude calls for care, since the usual real-valued normed-space interfaces are not interchangeable with an F -valued magnitude.

16.3 Scope of the executable checks

Two independent exact symbolic suites accompany this report, preserved unchanged in `code/` with their recorded outputs in `data/`; see appendix D for the inventory and for their independently reproduced results. Both use exact SymPy arithmetic on indeterminates and rational numbers, with no floating-point sampling and no network access.

The first suite checks the Hamilton table, generic associativity and norm identities, the complex matrix representation, Rodrigues' formula, the derivation reconstruction formula equation (12.4), the polynomial example Example 5.6, a Cayley unit, truncated noncommutative exponential, logarithm and BCH identities, the two-scale matrix of Example 13.4 through order t^9 , and the Fueter polynomial identities. Its recorded run passed all 69 check groups, comprising 252 scalar identities. The second suite checks generic associativity in twelve scalar variables, conjugation reversing a generic product, norm-square multiplicativity, the two-sided inverse, the commutator as twice the cross product, matrix representation multiplicativity, determinant and adjoint, unnormalized Rodrigues, the stereographic chart's exact unit norm, the full square-map Jacobian determinant of Proposition 15.2, the parallel/perpendicular inverse of the square linearization, the root asymmetry of Example 5.6 with its central norm polynomial, the interspersed no-root example of Warning 5.7, $\log(\exp x) = x$ and $\exp(\log(1 + x)) = 1 + x$ through degree six, BCH through degree four, the adjoint series through degree five, the graded group commutator leading term $2t^3\mathbf{k}$, the displaced square root equation (15.3) through degree seven, the radial exponential differential determinant equation (11.2), and $\mathcal{D}\Delta_4(q^n) = 0$ for $n = 0, \dots, 7$. Its recorded run passed all 31 named checks.

These checks detect algebraic sign errors, misplaced factors and expansion mistakes in the displayed examples. They do *not* verify set or class foundations, the Hahn support lemma, real-closed-field transfer, the analytic hypotheses, or the Berarducci–Mantova construction, and they do not establish any general analytic or spectral theorem. The general proofs remain the mathematical arguments in this article, relying on the explicitly cited foundational results.

17 Limitations and non-claims

Every honest limitation recorded by either source manuscript is reproduced here. They are numbered for reference and none of them is weakened.

- (N1) This is not a refereed article, not a priority claim, and not a claim to solve a named published open problem. Of theorem 12.5 and Corollary 12.6 in particular: we make no claim that this formulation has never appeared in the literature.
- (N2) Literature inspection was targeted, not an exhaustive priority search. Classical quaternion results and existing scalar-surreal constructions are attributed rather than presented as new, and no first-publication or open-conjecture claim is made anywhere.
- (N3) Much of sections 3 to 5 is acknowledged to be classical quaternion algebra over a real closed field [7], written out to expose hypotheses; the slice framework and the Fueter transformation

have classical antecedents [8]; and the scalar phase normalization is due to Ehrlich and Kaplan [6]. The genuinely surreal content is the class/set boundary, support control, multiscale estimates, and the incompatibility results.

- (N4) No Lean implementation or formal verification of this article is claimed. The archives contain no purported Lean proof; both layerings in section 16.2 are proposed modules only, and we refuse to infer a theorem from a successful build reported elsewhere.
- (N5) The 69 symbolic check groups and 252 scalar identities of the first suite verify finite polynomial and rational identities and truncated formal expansions only. They do not construct the surreals, do not prove the Hahn support lemma, do not verify real-closed-field transfer, do not verify the Berarducci–Mantova construction, and do not establish any general analytic or spectral theorem.
- (N6) The 31 checks of the second suite are finite identity and truncation checks only — not a verification of theorems about proper classes, arbitrary real closed fields, or infinite well-ordered supports. Its truncation order is an explicit finite parameter, and no finite truncation is presented as a proof of an infinite-series identity.
- (N7) Neither source manuscript makes any claim about any empirical subject, and none is made here. There is no assertion about spacetime, relativity, quantum mechanics, gauge theory, renormalization, measurement, or physical spin. The physics-adjacent vocabulary that does appear is strictly mathematical: “spin” refers to the algebraic double cover of equation (4.6); “rotation” always means an element of $\mathrm{SO}(3, F)$ or $\mathrm{SO}(4, F)$ acting on $\mathrm{Im} \mathbb{H}_F$, never a physical motion; the “avoided crossing” of Example 13.5 is finite Hermitian 2×2 linear algebra; the “angular velocity” identity equation (12.7) is a derivation identity in the surreal coefficients and expressly not a time evolution or an existence theorem for trajectories; and “condition number” in Proposition 15.2 and theorem 15.4 is an exact valuation-theoretic statement, not a numerical heuristic.
- (N8) The two-to-one map $S^3(F) \rightarrow \mathrm{SO}(3, F)$ is stated only as a map on algebraic group points, *not* as a covering-space theorem for the fine topology. Neither presentation of the rational chart in Proposition 4.5 claims that $\mathrm{Sp}(1, F)$ is topologically a real three-sphere, and the polar decomposition equation (4.4) is algebraic and does not assert compactness of $S^3(\mathbf{No})$.
- (N9) The root theorem covers *only* one-sided polynomials with a central formal indeterminate and coefficients on the right. Arbitrary interspersed-coefficient expressions can have no root at all, by both counterexamples of Warning 5.7; $\mathbb{H}_{\mathbf{No}}$ is a noncommutative division algebra, not an algebraically closed field.
- (N10) All spectral theory here is finite-dimensional matrix algebra. No Hilbert-space, infinite-dimensional or proper-class spectral theorem is asserted, and no compactness of a surreal unit sphere is used.
- (N11) The determinantal singular-scale theorem and the positive Cauchy–Binet argument of the companion spectral-theory report [15] are commutative determinant arguments and cannot be transferred unchanged to quaternionic matrices; no quaternionic determinant version is claimed here (Warning 13.2). Commutative results are used only with their scalar or slice hypotheses established.
- (N12) For matrices only $w(AB) \geq w(A) + w(B)$ holds. The equality equation (6.4), valid for individual nonzero quaternions, must not be copied to matrices.

- (N13) The cluster and block multiscale spectral calculus is a *direction*, not a theorem: only one rank-one instance is proved (theorem 13.3 and Example 13.4), with no optimal block constant.
- (N14) $\text{Exp}_{\mathfrak{m}}$ gives a *bijection* between the imaginary infinitesimals and U_1 , *not* an isomorphism of groups: the transported group law is noncommutative and is BCH.
- (N15) The global radial exponential depends on a *chosen* scalar normalization, that of Ehrlich and Kaplan; it is not an everywhere Hahn-summed power series and not an additive-to-multiplicative homomorphism on the whole additive algebra, so its fibre over 1 is not a kernel. There is no preferred logarithm: on the negative central axis every $\log_{\mathbf{No}} \rho + \pi I$ works, and ordinary 2π periods give further logarithms elsewhere. The fibre computations of theorem 10.5 do not settle the robustness questions for alternative surcomplex exponentials posed in [6].
- (N16) The tube exponential $\text{Exp}_{\mathcal{T}}$ is not a Hahn sum of $q^n/n!$ and is not $\exp(q_0) \text{Exp}_{\mathfrak{m}}(\varepsilon)$ in general (Warning 10.2), and its domain excludes ωi .
- (N17) Corollary 12.6 forbids a global commuting exponential chain rule for the componentwise Berarducci–Mantova derivation $\partial_{\mathbb{H}}$ only. It is not a statement about the multiplicative behaviour of any exponential, it does not conflict with the Liouville-closed real differential-field results of [5], and it does not contradict the commutative surcomplex homomorphism claim of [17]; see section 2.1 and Remark 12.7.
- (N18) Surjectivity of $\partial_{\mathbb{H}}$ is not a solvability theorem for nonlinear quaternionic differential equations; it gives coefficientwise antiderivatives only.
- (N19) A set-sized Hahn workspace need not be closed under Gonshor’s exponential or under a chosen derivation. Such closure is an additional hypothesis wherever those operations are used, and localization (theorem 1.1) does not supply it, nor does it identify a workspace’s intrinsic topology with its subspace topology in $\mathbf{No}$.
- (N20) Positive valuation is a summability certificate, not cofinality: a valid Hahn identity need not be the limit of its partial sums in any valuation topology (Warning 7.2). The word “convergent” is never used as a substitute for a named construction.
- (N21) Fine differentiation does not produce ordinary continuous paths and is explicitly not used to justify a general inverse-function theorem; theorem 15.1 is the existence tool. A zero fine differential does not imply constancy (Example 7.5), and the full class does not become a Banach-space manifold.
- (N22) Coefficientwise contour integration is over *ordinary* contours in an ordinary common domain, not over fine-continuous surreal contours, and it gives no uniqueness for arbitrary functions on $\mathbb{H}_{\mathbf{No}}$. No unrestricted maximum principle and no global residue theorem on arbitrary surreal paths is asserted, and the identity principle is deliberately coefficientwise.
- (N23) Admissible-series and coherent-function statements require a *common* support or a *common* ordinary domain. Knowing $w(a_n) \geq 0$ pointwise is not a substitute, a family with no common domain is a different space (Example 14.8), and ordinary-quaternion and Hahn-valued coefficient classes must not be interchanged (Warning 14.3).
- (N24) The $\sigma > 2\kappa$ conditioned square-root threshold of theorem 15.4 is asserted as sufficient only, not as optimal for every perturbation direction.
- (N25) All algorithms are effective *relative to* an exact scalar oracle. Arbitrary surreal coefficients have no finite computable encoding; root finding needs factorization over the real closed scalar

field; infinite supports can have infinitely many exponents below a cutoff, so truncation is not generally finitely computable; and the unrestricted-symbolic-expression row of section 16.1 carries no decidability or evaluation guarantee.

- (N26) The Cayley–Dickson construction over **No** also produces an octonion algebra with positive sum-of-eight-squares norm and hence no zero divisors, but it is *alternative* rather than associative, and the centralizer, linear-factor and matrix proofs in this article cannot be copied without revisiting their uses of associativity. Later Cayley–Dickson algebras already have zero divisors over $\mathbb{R}$ [9], and those examples persist under scalar extension to **No**; infinite coefficients do not repair an algebraic zero-divisor identity. This is an observation, not an octonionic theory.
- (N27) The research directions of section 18 are explicitly open. None of them is claimed solved by the constructions above.
- (N28) The first source archive’s README lists a file `SHA256SUMS.txt` that is *absent* from the delivered tree. The checksums it would have contained are therefore unavailable, and no integrity claim is made on their basis.
- (N29) Each accompanying script writes its own result file beside itself by default, so re-running it in place destroys the delivered record. The independent reproduction described in appendix D was therefore performed on copies; a byte-identity check after an in-place run would compare two equally modified copies and prove nothing.

18 Research directions

Six directions are left open, three from each source manuscript. None is claimed solved.

Collision-stable spectral subspaces. The rank-one estimate theorem 13.3 can be developed into cluster-projector estimates with a gap between two eigenvalue groups and no gap within either group. The natural quaternionic proof solves an off-diagonal Sylvester equation after spectral diagonalization and estimates denominators by the intercluster gap; iterating across a hierarchy of valuations should produce a block-diagonal multiscale spectral calculus. Example 13.4 proves one rank-one instance, not that iterative theorem ((N13)).

Support-aware eigenspace perturbation through collisions. Repeated quaternionic eigenvalues call for a support-aware perturbation theory of eigenspaces and spectral projections. Theorem 13.1 supplies existence but not a uniform series for an arbitrarily chosen eigenbasis through a collision.

Finite analytic zero algebras over quaternion coefficients. The sphere remainder algorithm completely solves finite-degree one-sided polynomials. For common-domain analytic stems, a useful next target is a support-certified local preparation theorem tracking isolated and spherical zeros under infinitesimal perturbation. Such a theorem must specify whether it takes place in a fixed-domain ring, a common-domain germ ring, or a ring of unrelated coefficient germs: Example 14.8 shows why conflating these rings can invalidate the proof before noncommutativity even enters.

Deformation of zero classes from spheres to points. Quaternionic zero classes can change from whole spheres to isolated points under small coefficient perturbations. A deformation theory should retain the entire finite conjugacy-class algebra, not merely a list of central roots; the explicit remainder $XA + B$ of Lemma 5.2 and the singular Sylvester operator of Proposition 15.2 identify the data such a theory must control.

Differential extensions containing oscillations. Theorem 12.5 is a definite obstruction rather

than a vague warning: constant-frequency rotations are absent from the componentwise Berarducci–Mantova quaternionic differential algebra. One concrete direction is to adjoin solutions of $\partial_{\mathbb{H}}y = Iy$ with a controlled involution and norm and study what replaces the finite-element smallness property; another is to retain the surquaternions and choose a different derivation, making its constants and order compatibility explicit; a third is to use only variable derivatives of common-domain stems and never identify them with a field derivation. These are different mathematical choices, not interchangeable notations for a single calculus.

Comparison of alternative global radial phases. A comparison of global radial phase extensions should separate their shared infinitesimal Lie theory from their infinite-radius critical sets. The character family of [14, 6] makes clear that theorems 10.5 and 11.2 describe one chosen normalization; which of their features are robust across the family is open.

18.1 The resulting picture

Surquaternions form an associative division algebra with an ordered-field-valued modulus, algebraically closed slices, exact rotation geometry at every scale, and a robust finite spectral and polynomial theory. Their normal forms carry the *noncommutative* residue algebra $\mathbb{H}_{\mathbb{R}}$, while the value group records genuinely surreal scales. Positive-support calculus supplies precise infinitesimal exponentials, logarithms, perturbation expansions and implicit lifting, and ordinary coefficient analysis supplies coherent slice and Fueter operations.

The structures fit together provided their boundaries remain visible. The global phase of an infinite imaginary argument is not determined by a raw power series. Hahn summation is not a universal topological limit. The field derivation is not differentiation of an analytic variable, and neither is the fine differential or the Fueter operator. The no-oscillation theorem proves that at least one tempting identification of these structures is impossible, while the constructive results show how much useful theory remains once that identification is abandoned — and how much of it must be reproved rather than imported, because the algebra does not commute.

A Notation ledger and conventions

This table is the ledger of section 2 in full, extended to cover every object either source manuscript introduced.

Notation or construction	Exact convention
$F, \mathbf{No}, \mathbb{H}_{\mathbb{R}}$	A real closed ordered field; the surreal proper class; the ordinary quaternions over $\mathbb{R}$.
$\mathbb{H}_F, \mathbb{H}_{\mathbf{No}}$	Hamilton algebra over F , and over $\mathbf{No}$. Scalar coefficients are central.
$\bar{q}, N(q), q $	Quaternion conjugation (an anti-involution), reduced norm $q\bar{q}$, and its positive square root in the ordered center. Not real-valued and not floating point.
$\operatorname{Re} q, \operatorname{Im} q$	Central scalar part and pure imaginary part; not maps to ordinary real coefficients unless the input is ordinary.

Notation or construction	Exact convention
$\mathbb{S}(F)$	Imaginary square roots of -1 ; $F_I = F + FI$ is a complex slice, algebraically closed for real closed F .
t^γ	Conway monomial $\omega^{-\gamma}$, <i>not</i> an unqualified exponential power and not $\exp(-\gamma \log \omega)$.
K_Γ, D_Γ	$\mathbb{R}((t^\Gamma))$ and $\mathbb{H}_{K_\Gamma} \simeq \mathbb{H}_{\mathbb{R}}((t^\Gamma))$, for a divisible set-sized ordered subgroup $\Gamma \subseteq (\mathbf{No}, +)$.
$\text{supp}(q), w(q), \text{lc}(q)$	Well-ordered set of increasing exponents; least exponent, with $w(0) = +\infty$; leading ordinary quaternion coefficient. Source 11 writes ν for w .
$\mathcal{O}_H, \mathfrak{m}_H, \text{st}$	Finite coordinates, infinitesimal coordinates (a two-sided maximal ideal), and coordinate-zero extraction onto the noncommutative $\mathbb{H}_{\mathbb{R}}$. In a workspace these are written $\mathcal{O}_D, \mathfrak{m}_D$ and $\mathcal{O}_K, \mathfrak{m}_K$ for scalars.
S^*	All finite sums from a positive well-ordered support S , including 0.
Hahn-summable	Set-indexed, well-ordered union of supports, each exponent in finitely many members; summed coefficientwise. Never a limit of partial sums.
$P \star Q, P^c, N_P$	Product with a central indeterminate and right-coefficient evaluation, <i>not</i> pointwise multiplication; coefficient conjugation; central symmetrization $P \star P^c$, also written P^s in source 11.
$\Delta_{a,r}, [a, r]$	$X^2 - 2aX + a^2 + r^2$; the conjugacy sphere $a + r\mathbb{S}(F)$.
$S^3(F), \text{Sp}(1, F), U, U_1$	The modulus-one group; the finite units of modulus one; its principal subgroup $U \cap (1 + \mathfrak{m}_H)$.
$\text{Exp}_{\mathfrak{m}}, \text{Log}_{\mathfrak{m}}$	Infinitesimal Hahn exponential and logarithm at positive valuation; $\text{Exp}_{\mathfrak{m}}$ in source 10 and $\text{exp}_{\mathbb{H}}$ in source 11.
$\text{Exp}_{\mathcal{T}}$	Finite-angle tube exponential on $\mathcal{T}$: Gonshor scalar exponential times canonical finite angle; finite imaginary part, unrestricted scalar part; Exp_0 in source 10.
$\text{exp}_{\mathbf{No}}, \text{log}_{\mathbf{No}}$	Gonshor's scalar exponential on $\mathbf{No}$ and its inverse.
$\text{fin } x, \mathbf{Oz}$	Real-plus-infinitesimal part of the scalar normal form (additive, not multiplicative, $\text{fin } \omega = 0$); the omnific integers $\mathbf{No}_{\text{PI}} + \mathbb{Z}$.
Sin, Cos	Global integer-part-normalized phase, $\text{sin}(\text{fin } x)$ and $\text{cos}(\text{fin } x)$; distinct from the finite-angle sin, cos of equation (9.1).
Exp_{rad}	Global radial quaternion exponential equation (10.6) using the Ehrlich–Kaplan normalization; Exp_{EK} in source 10 and Exp_{can} in source 11.
θ, η, B	Finite angle; infinitesimal phase residue; accumulated phase. The three senses fixed in [16]; no fourth sense is used.
$\partial, \partial_{\mathbb{H}}$	Berarducci–Mantova scalar derivation with $\partial\omega = 1$; its componentwise extension to $\mathbb{H}_{\mathbf{No}}$ (D in source 10, ∂_H in source 11).
$\delta, \delta_{\mathbb{H}}, \text{ad}_w$	An arbitrary scalar field derivation; its componentwise extension (D_0 in source 10); the inner part of a general derivation.
∂_{sl}	Coefficientwise slice (formal) derivative of a right-coefficient series; not the full differential.
df_q	Fine ($\mathbf{No}$ -linear Fréchet) differential of Definition 11.1; not a Banach derivative.

Notation or construction	Exact convention
$\partial_a, \partial_b, \partial_r$	Variable derivatives of ordinary coefficient functions and their Taylor extensions; never $\partial_{\mathbb{H}}$.
ι	Auxiliary central complex unit for stems; distinct from every quaternion unit.
$\mathcal{D}, \Delta_4$	Left Fueter operator and four-coordinate variable Laplacian.
$\int^H$	Ordinary contour integration of individual Hahn coefficient functions, with a uniform support certificate; not a fine-topological limit of Riemann sums.
L_{q_0}	The Sylvester operator $h \mapsto q_0 h + h q_0$ of Proposition 15.2.
Fine topology	Balls of every positive surreal radius on the full class; distinct from a workspace's intrinsic valuation topology and from the coarse standard-part topology (Remark 7.4).

B Dependencies beyond the displayed proofs

Results	Main dependencies beyond their displayed proofs
Division algebra, centralizers, conjugacy	Ordered-field algebra; positive square roots where normalization is used.
Polynomial root algorithm, spectral theorem	Real closedness of F and algebraic closedness of $F[\mathbf{i}]$.
Surreal realization and localization	Conway normal-form theorem and the real-closed Hahn-field theorem [1, 3, 6].
Hahn inverse, principal logarithm, BCH, adjoint series, implicit lifting	Positive-support Neumann lemma (Lemma 7.1, via Higman's finite-word lemma) and coefficientwise strong re-grouping.
Finite angles and global quaternion exponential	Gonshor's real exponential and the Ehrlich–Kaplan global circular character [2, 6]; the scalar statements are inherited from [14].
No-oscillation theorem and chain-rule corollary	Berarducci–Mantova constant field, smallness, and $\partial\omega = 1$ [5]; the scalar precursor is inherited from [16].
Cauchy and Fueter results	Ordinary coefficient analysis on one common domain, plus Hahn Taylor support certificates; classical slice theory [8].
Fixed-dimension algebraic transfer	First-order real-closed-field transfer, subject to expressibility [10].

C Worked exact calculations for reuse

C.1 Conjugating two imaginary units

For $I = \mathbf{i}$ and $J = \mathbf{j}$, the conjugating element of theorem 4.1 is $h = 1 - \mathbf{j}\mathbf{i} = 1 + \mathbf{k}$. Normalization gives $u = (1 + \mathbf{k})/\sqrt{2}$, and

$$u\mathbf{i}u^{-1} = \mathbf{j}, \quad u\mathbf{j}u^{-1} = -\mathbf{i}, \quad u\mathbf{k}u^{-1} = \mathbf{k}.$$

Replacing the constant direction by a surreal one does not change the formula: for example $J = (\mathbf{i} + t\mathbf{j})/\sqrt{1+t^2}$ produces an exact infinitesimal rotation through the same construction, with no numerically approximated angle.

C.2 An inverse and its error certificate

For $x = t\mathbf{i}$ one has $x^2 = -t^2$, so

$$(1+x)^{-1} = \frac{1-t\mathbf{i}}{1+t^2} = 1 - t\mathbf{i} - t^2 + t^3\mathbf{i} + t^4 - \dots$$

The difference from the first $M+1$ powers $\sum_{n=0}^M (-x)^n$ is exactly $(-x)^{M+1}(1+x)^{-1}$, of valuation $M+1$. This is an equality and a certificate, even in the full fine topology where the displayed partial sums do not converge as an ordinary sequence (Warning 7.2).

C.3 The first noncommutative correction

For $x = t\mathbf{i}$ and $y = t\mathbf{j}$ the degree-two term of $\text{BCH}(x, y)$ is $t^2\mathbf{k}$, and its degree-three term is

$$\frac{1}{12}([t\mathbf{i}, [t\mathbf{i}, t\mathbf{j}]] + [t\mathbf{j}, [t\mathbf{j}, t\mathbf{i}]]) = -\frac{t^3}{3}(\mathbf{i} + \mathbf{j}),$$

so

$$\text{BCH}(t\mathbf{i}, t\mathbf{j}) = t(\mathbf{i} + \mathbf{j}) + t^2\mathbf{k} - \frac{t^3}{3}(\mathbf{i} + \mathbf{j}) + \text{terms of valuation at least 4}.$$

A scalar exponential implementation that simply adds the exponents would lose the entire $t^2\mathbf{k}$ rotation. Both verification suites check this coefficient calculation by truncated noncommutative multiplication, independently of the displayed bracket simplification; see also the unequal-scale form in Example 8.3.

C.4 The two-scale spectrum at integer exponents

For $\alpha = 1$, $\beta = 2$, equation (13.6) specializes to

$$\lambda_- = \frac{t - t\sqrt{1+4t^2}}{2} = -t^3 + t^5 - 2t^7 + 5t^9 + \dots, \quad \lambda_+ = t + t^3 - t^5 + 2t^7 - 5t^9 + \dots,$$

and the off-diagonal entry of its lower projector is

$$-\frac{t\mathbf{i}}{\sqrt{1+4t^2}} = -t\mathbf{i} + 2t^3\mathbf{i} - 6t^5\mathbf{i} + 20t^7\mathbf{i} + \dots$$

The ordinary standard-part matrix is zero, yet a meaningful eigenspace is resolved at order t , with its perturbation explicitly visible. This is one reason ordinary floating-point substitution at a small positive real t should be treated as an illustration rather than as the definition of the surreal problem.

C.5 The radial exponential at an infinite radius

The two computations to keep side by side are

$$\mathrm{Exp}_{\mathrm{rad}}(\omega \mathbf{i} + \pi \mathbf{j}) = 1 + \frac{\pi^2}{2\omega} \mathbf{i} + O(\omega^{-2}), \quad \mathrm{Exp}_{\mathrm{rad}}(\omega \mathbf{i}) \mathrm{Exp}_{\mathrm{rad}}(\pi \mathbf{j}) = -1,$$

from Example 10.6, together with the critical-point expansion

$$\mathrm{Exp}_{\mathrm{rad}}(\omega I + \eta J) = 1 + \frac{\eta^2}{2\omega} I + \frac{\eta^3}{2\omega^2} J + \cdots \quad (J \perp I)$$

of Example 11.3, whose vanishing first-order transverse term is the rank-two degeneracy of equation (11.2) at $r = \omega \in \pi \mathbf{Oz}$. The recorded suites check the determinant formula symbolically.

C.6 The displaced square root and its recursion

Equation (15.3) gives $q = 1 + \mathbf{i} + \frac{t}{2}\mathbf{j} + \frac{t^2}{16}(1 - \mathbf{i}) - \frac{t^3}{32}\mathbf{j} + O(t^4)$ as the unique root of $q^2 = (1 + \mathbf{i})^2 + t\mathbf{j}$ near $1 + \mathbf{i}$, obtained from the exact Sylvester recursion of section 15.1. The coefficients $\mathbf{j}/2$, $(1 - \mathbf{i})/16$ and $-\mathbf{j}/32$ are checked symbolically through degree seven.

D Reproducibility, provenance, and archive inventory

This source is a standalone L^AT_EX article with an embedded bibliography, requiring no external figures, bibliography database or proprietary fonts. Build it with

```
latexmk -pdf -interaction=nonstopmode article.tex
```

and inspect `article.log` for errors, undefined references, undefined citations, multiply-defined labels and duplicate destinations before accepting the output.

Source manuscripts. This report is the merge of two independently written manuscripts, *noncommutative-hahn-obstructions* and *radial-exponential-support-lifting*, neither of which is reproduced here. The first was delivered as a 34-page article with 27 numbered statements; the second as a 32-page article. The disposition of every result either of them reached is recorded in the provenance material that remains: the scope and provenance of section 1, the ledger of section 2, and the limitations of section 17.

Code and data. Two independent suites are preserved unchanged in `code/`, with their recorded outputs and metadata in `data/`. Both pin `sympy==1.14.0`. The first suite’s record states 69 of 69 check groups passed, 0 failed, 252 scalar identities, with no failing entry, under Python 3.13.5 and SymPy 1.14.0; its build metadata records 34 pages, 27 theorem-type statements, 27 proof environments, zero L^AT_EX errors, warnings, overfull or underfull boxes, zero unresolved reference markers, and `formal_theorem_proof_verification: false`. The second suite’s record states 31 of 31 checks passed with `all_passed: true`, under the same versions, and describes itself as “exact finite algebra and truncated-series checks, not formal theorem verification”.

Independent reproduction. Both suites were re-run independently on *copies* of the trees, under Python 3.14.4 with SymPy 1.14.0, and both exited zero, reporting 252 identities in 69 groups and 31 checks respectively. Running on copies is not a formality: each script writes its own result file beside itself by default, so an in-place re-run overwrites the delivered evidence, and a byte-identity check performed afterwards would compare two equally modified copies and prove nothing ((N29)). Nothing under `code/` or `data/` was modified in producing this report.

Provenance and status. Both source manuscripts recorded the same repository snapshot, 39f2be6667ade51bca2b45daa47e289d69c09764, inspected 21 September 2026, as orientation only [12, 13, 14]; the first records `repository_modified: false` and `repository_build_run: false` in its manifest. Neither treated the repository’s build as a formal verification of its theorems, and neither claimed to have added or compiled any Lean module. Both are AI-assisted research drafts with written proofs and finite symbolic checks; neither is refereed or machine-checked, and this merged report inherits exactly that status. Every limitation of either manuscript is reproduced in section 17.

Reading routes. For an algebra-first reading follow sections 3 to 6 and then section 16. For infinitesimal geometry follow sections 6 to 9. For the exponential at infinite arguments read sections 2 and 9 to 11 together: omitting the scalar normalization of section 10.2 changes the subject. For analysis and formalization read sections 7 and 14 to 16 with the four operators of Convention 2.2 kept apart.

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